

From the last form (valid only if  $r_3 \neq 0$ ), we see that the ratio  $r_2/r_3$  is a zero of  $z^2 + pz - q = 0$  (note the plus sign on the  $z$  term), and if the original quadratic  $Q(z) = z^2 - pz - q$  has complex zeros, then  $D \neq 0$ . If  $r_2 = r_3 = 0$ , then we are “near” a possible multiple zero which we will investigate later. The possibility that  $q = 0$  and  $r_2 = 0$  is also unlikely [though we shall start with  $q = 0$  as our starting approximation for  $Q(z)$ ] since we have removed all zeros at the origin, which implies  $q \neq 0$ .

The modification of Newton’s method (Sec. 4.6) uses a metric  $|P(z)|$ . We shall at this point use the equivalent, but more convenient, metric

$$|P(z)|^2$$

evaluated at the point where  $Q(z) = z^2 - pz - q = 0$ . Thus

$$\begin{aligned} |P(z)|^2 &= (r_1 z + r_0)(r_1 \bar{z} + r_0) \\ &= r_1^2(-q) + r_1 r_0 p + r_0^2 \end{aligned}$$

Experience shows that using this modification of Newton’s method, Bairstow’s method will converge *even* for multiple factors and cannot wander or fail to converge.

When we find a quadratic factor, we need to check that its zeros are not near the imaginary axis (we have in the removal of the real zeros effectively checked that they are not near the real axis.) If  $p^2$  is very much less than  $-4q$  (which is a positive number), then we need to consider whether or not the quadratic factor  $z^2 - q$  would be acceptable within roundoff, and if it is, then we prefer it.

## 6.8 CONVERGENCE OF BAIRSTOW’S METHOD

The unmodified Bairstow method is famous for not converging, but the simple modification of Newton’s method removes most of this trouble. It is worth examining further to see why we do not get into trouble in this application of Newton’s method in two variables—in particular why a *local* minimum does not trap us.

In the theory of functions of a complex variable there is a minimum modulus theorem (much like the maximum modulus theorem) which says that if at a point  $z_k$  in the  $z$  plane the function  $f(z) \neq 0$  (and  $f(z)$  is not a constant), then there are values of  $f(z)$  in the immediate neighborhood of  $z_k$  such that

$$|f(z)| < |f(z_k)|$$

We are, of course, not in the  $z$  plane, but as discussed in Sec. 6.6, we are operating in the  $pq$  plane inside the parabola and not “near” the edge. Therefore there is

a local one-to-one mapping of points in one plane onto points in the other, and there are points in the  $pq$  plane where the function is lower—in this problem there can be no local minima other than zero.

When we solve the Bairstow equations for the next step (except in the fantastically unlikely case of exact cancellation of roundoff effects), we will find that there are terms in the polynomial that will produce a step, possibly small, that takes us away from any local saddle points (the two-dimensional equivalent of a local minimum trouble in one variable), and we will not get stuck, unable to go to a lower point. It was partly for this reason that we reduced the polynomial to one which had no common factor of the exponents of the nonzero terms.

## 6.9 MULTIPLE ZEROS

We have developed the theory for the linear and quadratic factors so that they parallel each other. Thus there is little more to be said about finding multiple quadratic factors except to say we do the analogous steps.

We should not become too excited about how accurate in fact are various multiple or close zeros. Consider, for example, the two functions

$$f_1 = (z + 1)^8 = z^8 + 8z^7 + 28z^6 + 56z^5 + 70z^4 + 56z^3 + 28z^2 + 8z + 1$$

$$f_2 = (z + 1)^8 + \left(\frac{1}{10}\right)^8 = z^8 + \cdots + 1 + \left(\frac{1}{10}\right)^8$$

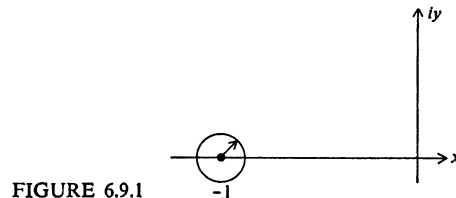


FIGURE 6.9.1

To eight decimal places they are the same; yet the first has an eightfold zero at  $z = -1$ , while the other has its zeros on a circle of radius  $1/10$  about  $z = -1$ . Evidently near multiple zeros it is difficult to know what is really happening—but often in practice it does not matter. If it does, then probably the problem was posed wrong! But for computing purposes we must report them as multiple zeros *if* we are to avoid subsequent roundoff troubles, and the multiple zero can be located reasonably accurately.

# 7

## LINEAR EQUATIONS AND MATRIX INVERSION

### 7.1 INTRODUCTION

The problems of simultaneous linear equations and matrix inversion have been exhaustively studied, and yet our knowledge is not all that we wish it were. Indeed, the very volume of papers indicates the difficulties of the two problems. We shall begin by presenting the simpler, rather widely accepted views of the matter and gradually point out the deficiencies in our theories that arise in some circumstances. In spite of the many objections that we shall raise for which there are at present only a few partial answers, the present methods often provide satisfactory results. Many of the points we wish to make will be illustrated by simple examples (it does not take a big machine to study and do research in numerical analysis), but very little is known as to how typical the examples are in practice.

<sup>1</sup> (Johann) Carl Friedrich Gauss (1777–1855).

we divide the first equation by  $a_{1,1}$  and then subtract  $a_{2,1}$  times this first result from the second equation,  $a_{3,1}$  times the initial result from the third, etc., until we have  $N - 1$  equations in the  $N - 1$  variables  $x_2, x_3, \dots, x_N$ . Using these  $N - 1$  equations, we eliminate  $x_2$  in the same way, leaving  $N - 2$  equations in  $x_3, \dots, x_N$ . Repeating this process a total of  $N - 1$  times, we finally come down to one equation in the variable  $x_N$ .

In the *back solution* we solve the last equation for  $x_N$ , put this in the next-to-last equation to get  $x_{N-1}$ , put these two into the third from the last equation to get  $x_{N-2}, \dots$ , until we have all the unknowns (determined in the order  $x_N, x_{N-1}, \dots, x_1$ ).

We now have our solution. The amount of arithmetic is roughly<sup>1</sup>

$$N^2 + (N - 1)^2 + (N - 2)^2 + (N - 3)^2 + \cdots + 1 = \frac{N(N + 1)}{2} \frac{2N + 1}{3}$$

additions and the same number of multiplications—a total of approximately  $2N^3/3$  operations in the forward part of the solution. The back solution has of the order of  $N^2$  operations, and for large  $N$  this is not significant.

Sometimes the value of the determinant of the system of equations is needed. When we divide an equation by its leading coefficient, we alter the value of the determinant by this amount. But when we multiply an equation by a constant *and* add it to another equation, we do not alter the value of the determinant. Thus the value of the determinant is the product of the  $N$  divisors we use.

## PROBLEMS

Solve and find the determinant value:

7.2.1  $3x + 6y + 9z = 39$

$$2x + 5y - 2z = 3$$

$$x + 3y - z = 2$$

7.2.2  $x + y + z = 6$

$$2x + 3y + z = 1$$

$$x - y + z = 3$$

7.2.3  $x - y + z = 1$

$$2x + 2y - z = 3$$

$$3x - 3y + z = 1$$

7.2.4  $ax + by = e$

$$cx + dy = f$$

<sup>1</sup> See Sec. 11.3 for the summation formula.

### 7.3 PIVOTING

If the solution of simultaneous linear equations is this simple, then why give all this attention to the problem? The reason is simply that in the above description we have overlooked many important details.

In the first place we ignored the possibility that one of the  $N$  coefficients that we were dividing by might have been zero. The answer is, of course, to pick another nonzero coefficient, but which one? The customary description says to pick as our *pivot* (as it is called) the largest in the column of the unknown you are eliminating. This is called *partial pivoting*.

This in turn raises the question of in which order the unknowns should be eliminated. Why the order  $x_1, x_2, \dots, x_N$ ? If instead of looking only in the next column we look at the array of all the coefficients  $a_{i,j}$  and pick the largest as the pivot, then this is called *complete pivoting*. It is generally agreed that while complete pivoting is not worth the very large effort necessary to find the largest of all those left at each stage, partial pivoting is worth the effort.

Partial pivoting can be regarded as a reordering of the equations, though this should not be done in the machine since it is apt to confuse the user of the results if he ever wants to see how the elimination went in his problem. Complete pivoting can be regarded as reordering both the equations *and* the subscripts of the unknowns.

The conventional argument for using the largest element as the pivot goes as follows. Selecting the largest element means that at each stage when it is combined with another equation, the pivoting equation will be multiplied by a number less than 1 in size, suggesting that the roundoff will therefore be less. But this is not true in a very real sense. Let  $L_i$  be the  $i$ th equation and  $c$  be the multiplier. If we eliminate one way, we get (symbolically)

$$cL_i + L_j = L'_j = \text{the new equation}$$

while if we eliminate using the *same* two equations but in the reverse way, we get

$$L_i + \frac{1}{c}L_j = \frac{1}{c}L'_j = \text{the new equation}$$

It should be obvious that these two equations have the same relative floating-point errors but different fixed-point errors. Thus the widely used argument that we should use large pivots *because* they reduce the roundoff is not sound for floating-point computation.

## PROBLEMS

Solve, using pivoting:

7.3.1  $x + y + z = 6$

$2x - y + z = 3$

$3x + 2y - z = 4$

7.3.2 Prob. 7.2.3

## 7.4 GAUSS-JORDAN ELIMINATION

If instead of the back solution we use the last equation to eliminate  $x_N$  in the top  $N - 1$  equations and then use  $x_{N-1}$  in the next-to-last equation to eliminate all the  $x_{N-1}$ 's, etc., we will come to a diagonal system of equations (Fig. 7.4.1) with

$$\begin{array}{ccccccc}
 x & x & x & \dots & x & x & x & = & x \\
 & x & x & \dots & x & x & x & = & x \\
 & & x & \dots & x & x & x & = & x \\
 & & & & x & x & x & = & x \\
 & & & & & x & x & = & x \\
 & & & & & & x & = & x
 \end{array}$$

FIGURE 7.4.1

the solution explicitly given. This is known as the *Gauss-Jordan method* of solution. It is not recommended in practice, but we shall need the idea a little later. Sometimes the Gauss-Jordan method is described as eliminating each unknown from the earlier equations just after the unknown is eliminated from the equations below it. This, clearly, requires more arithmetic and is not recommended.

## 7.5 SCALING

The method of pivoting used words like “pick the largest element,” but if instead of writing down an equation we wrote down  $c$  times the equation, we would quite possibly get a different pivot. Clearly the algorithm is not invariant as it stands; what we do will depend in many problems on the haphazard way the equations were written down. This suggests that some initial scaling is necessary

before we begin the solution process. Scaling can be regarded as trying to make the algorithm invariant.

We can clearly multiply the  $i$ th equation by any nonzero constant  $r_i$  we please and mathematically still have the same set of equations, and we can replace the  $j$ th variable by  $c_j x_j$ , which effectively multiplies the  $j$ th column by  $c_j$ . And we could, if we pleased, multiply the right-hand sides  $b_i$  by a constant, as well as multiply all the numbers in the problem by a constant  $M$ . These give the transformations that we intuitively feel leave the system of equations the same and for which we wish our algorithm to be invariant.

If we were to pick the same pivots each time (as judged by the subscripts  $i, j$ ), then the algorithm would be invariant under all transformations of the same problem, though it will take a little thought on the reader's part to follow the elimination process and the back solution processes through to convince himself of this. The  $r_i$  merely affect the multipliers used to do the elimination, the  $c_j$  change all the numbers in the  $j$ th column by the same factor, the right-hand sides similarly are merely scaled by a multiplication constant, and the multiplier  $M$  has no real effect.

We face, therefore, the problem of making some sense of the words "pick the largest element in the next column as the next pivot." It is conventional to say that the transformations should be used to scale the rows so that the maximum in each row is around 1 in size and then we scale the columns similarly (or else scale the columns and then the rows). Sometimes it is observed that we should not actually perform the scaling, as this will cause unnecessary roundoff, but merely store away the scaling factors and use them when we come to "pick the largest . . ."

As we shall later show, scaling by rows and then columns, or scaling by columns and then rows, does not lead to the same result; indeed, one way may lead to disaster while the other does not. Thus the usual advice about scaling is both superficial and misleading.

If this is so, then what about the whole method of Gaussian elimination? Before we panic, it should be observed that many times the problem comes in a form where the equations have been "naturally scaled" by the way they came about, especially in equations that arise in mathematical problems. But there are times when the equations do indeed need some preliminary scaling before starting the elimination process. Section 7.6 discusses one possible approach to scaling that produces a unique result and has the property that any time you alter the system by any of the transformations:

Interchange of rows

Multiplication of a row by a nonzero constant  $r_i$



Interchange of columns

Multiplication of a column (including the right-hand side) by a nonzero constant  $c_i$

Multiplication of everything by a constant  $M$

the scaling method will, within roundoff, remove the transformation effects, thus producing the basis for an invariant algorithm.

## 7.6 INVARIANT SCALING—ANALYSIS OF VARIANCE

The method of analysis of variance in statistics provides a natural tool for the invariant scaling of a set of simultaneous linear equations.

We suppose, first, that all the  $a_{i,j} \neq 0$  and set

$$|a_{i,j}| = 2^{d_{i,j}}$$

We imagine multiplying each row by  $2^{r_i}$  ( $i = 1, 2, \dots, N$ ), each column (including the right-hand side) by  $2^{c_j}$  ( $j = 1, 2, \dots, N + 1$ ), and everything by  $2^M$ . The exponent in position  $i, j$  is now

$$d_{i,j} + r_i + c_j + M$$

If we now minimize the sum of the squares of all the exponents, we will find values such that, in the least-squares sense, the exponents will be as close to zero as they can be made to be; that is, the variance of the exponents is minimized.

Therefore we minimize

$$m = \sum_{j=1}^{N+1} \sum_{i=1}^N (d_{i,j} + r_i + c_j + M)^2$$

Differentiating with respect to each of the unknowns  $r_i$  and  $c_j$ , we get

$$\begin{aligned} \frac{\partial m}{\partial r_i} &= 2 \sum_{j=1}^{N+1} (d_{i,j} + r_i + c_j + M) = 0 \quad i = 1, 2, \dots, N \\ \frac{\partial m}{\partial c_j} &= 2 \sum_{i=1}^N (d_{i,j} + r_i + c_j + M) = 0 \quad j = 1, 2, \dots, N + 1 \end{aligned}$$

We have more than enough parameters, and so we arbitrarily set

$$M = \frac{-1}{N(N+1)} \sum_{j=1}^{N+1} \sum_{i=1}^N d_{i,j}$$

The solution to these equations is

$$r_i = \frac{-1}{N+1} \sum_{j=1}^{N+1} (d_{i,j} + M) \quad \text{hence } \sum_{i=1}^N r_i = 0$$

$$c_j = \frac{-1}{N} \sum_{i=1}^N (d_{i,j} + M) \quad \text{hence } \sum_{j=1}^{N+1} c_j = 0$$

as can be seen by substituting these into the derivative equations

$$\sum_{j=1}^{N+1} [d_{i,j} - (d_{i,j} + M) + 0 + M] = 0$$

$$\sum_{i=1}^N [d_{i,j} + 0 - (d_{i,j} + M) + M] = 0$$

Thus  $r_i$  and  $c_j$  are the appropriate negative averages of

$$d_{i,j} + M$$

and  $M$  is the negative of the average over all the  $d_{i,j}$ .

This approach minimizes the variance of the scaled exponents, which seems like a good thing to do. However, we are unable to connect this property directly with the subsequent elimination process, or at present with any other method of solution, so that the method is merely suggestive and plausible.

If any of the coefficients of the original system of equations were zero, then the corresponding value of  $d_{i,j}$  would be minus infinity, and we exclude them as “missing data.” With data missing the solution of the system is harder to find. We refer the reader who wishes to pursue this unconventional scaling method to the standard books on the topic of analysis of variance. We note, however, that we are merely looking for a reasonable solution to these equations to get some way of reasonably scaling the original system. Note that we do not need to get integral solutions to minimize the roundoff due to scaling, because we intend not to actually scale the equations but only to use these numbers to pick the pivots in the original system of equations

## 7.7 RANK

During Gaussian elimination on a system of linear equations it may happen that at some point all the rest of the coefficients in a column are zero. This means that the rank of the matrix of coefficients is not  $N$  but lower. Indeed, in principle we may find at several stages that we are blocked by a column of zeros. Equally likely we may find a row, or rows, of zeros blocking us.

In the case of finding a row of zeros this clearly means that the linear combination of the earlier rows used is equal to the row of all zeros; it means that there is a linear dependence and that the rank is not  $N$ .

Most of the time the proposer of the problem believed that the system of equations was of full rank and that it would determine a unique solution. The fact that the system is not of full rank does not end the problem, since the proposer needs to know what the dependence is in order to go back and reformulate his problem. How can we find the dependence? We could follow the elimination process to find the linear combination that produced it. Alternatively, we can drop one of the earlier rows and try again; if at the same point we again find all zeros, then we know that the dropped row is not part of the dependence. But if we do not find all zeros, then it is part of the dependence. Dropping one row at a time, we can find the minimal linearly dependent set. Then it is a matter of finding the actual coefficients of these equations, which can be done by keeping track of the coefficients used in the elimination process.

It is a central theorem of linear algebra that the row and column ranks are the same, which means that if we find a row dependence, then there must be a corresponding column dependence, and vice versa. Evidently we can operate on the columns as we did the rows to find a column dependence.

Mathematics often does not make distinctions that some applications require. For example, Euclidean geometry does not distinguish between left-handed and right-handed triangles: given three sides of one triangle equal to three sides of another triangle, the triangles are congruent (see Fig. 7.7.1). But for many applications (such as wearing gloves, studying some chemical compounds, etc.) it is necessary to distinguish between the two different orientations. For these applications Euclidean geometry must be altered a bit.

Similarly for simultaneous equations: though in principle the row and column ranks are the same, in practice the meaning to be attributed to one may be different from the other. In practice a simple row dependence, say two rows are the same, is quite different from two columns being the same.

| <i>Row dependence</i>                                                                                                                                                                                                                                                                                                                                                                                | <i>Column dependence</i>                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $  \begin{array}{rcl}  2x + 4y + 6z & = & 10 \\  x + 2y + 3z & = & 5 \\  x - y - z & = & -7 \\  x + 2y + 3z & = & 5 \\  0y + 0z & = & 0 \\  -3y - 4z & = & -12  \end{array}  $ <div style="position: relative; height: 100px;"> <div style="position: absolute; top: 0; left: 20px;"> <p>pivot</p> </div> <div style="position: absolute; bottom: 0; left: 20px;"> <p>row of zeros</p> </div> </div> | $  \begin{array}{rcl}  2x + 4y + 6z & = & 10 \\  x + 2y - z & = & 4 \\  x + 2y + 7z & = & 6 \\  x + 2y + 3z & = & 5 \\  0y - 4z & = & -1 \\  0y + 4z & = & 1  \end{array}  $ <div style="position: relative; height: 100px;"> <div style="position: absolute; top: 0; left: 20px;"> <p>pivot</p> </div> <div style="position: absolute; bottom: 0; left: 20px;"> <p>column of zeros</p> </div> </div> |

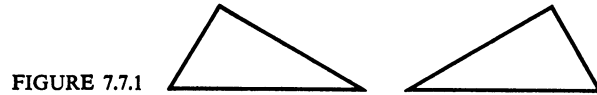


FIGURE 7.7.1

This difference means that in partial pivoting when we find a zero element, we should at that point check to see if the whole row is zeros (we will automatically find the column of zeros if that is the case). When we find a dependence, we should isolate it as best we can and exhibit it to the user. We should, probably, continue to search for further dependencies before stopping.

Sometimes it is necessary to carry out the solution of a system of linear equations when the rank is less than  $N$ . We cite without proof the main result for this. If the rank of the system is  $r < N$ , then some  $N - r$  of the variables may be transferred to the right-hand side and the system solved in terms of them as if they were parameters. If the rank of the augmented matrix is the same as the rank of the matrix, then there is an  $(N - r)$ -parameter solution; otherwise the equations are inconsistent.

### PROBLEMS

7.7.1 Find the rank and linear dependence.

$$x + y + z = 2$$

$$2x + 4y - 6z = 8$$

$$2x + 3y - 2z = 6$$

7.7.2 Solve the system

$$x + y = a$$

$$2x + 2y = 2a$$

7.7.3 Find the linear dependence, both row and column, of:

$$x + 3y - z = 5$$

$$2x - y - 2z = 7$$

$$4x + 5y - 4z = 17$$

### 7.8 ILL-CONDITIONED SYSTEMS

All the above has been carried out in terms of exact numbers. The floating-point numbers we use will generally not produce rows or columns of exact zeros, and we will have trouble determining the rank. Indeed, it may well be that the row and column ranks are not the same as judged by our tests for rank. How to resolve this kind of trouble is not very well understood at present.

Given the system

$$x + \frac{1}{3}y = 1$$

$$2x + \frac{2}{3}y = 2$$

The rank is 1.

Put the system in a computer:

$$\begin{array}{l} \text{Note coefficients} \\ \downarrow \\ 0.100 \times 10^1 x + 0.333 \times 10^0 y = 0.100 \times 10^1 \\ \downarrow \\ 0.200 \times 10^1 x + 0.667 \times 10^0 y = 0.200 \times 10^1 \end{array}$$

Now the rank is 2, and we can solve the system uniquely!

$$0.100 \times 10^{-2} y = 0$$

$$\begin{cases} y = 0 \\ x = 1 \end{cases}$$

Our trouble with determining the rank plus our trouble with initial scaling to give pivoting some meaning indicate that we need to find something to replace these two ideas. This needed idea is “ill-conditioning,” and as used in the literature it is ill-defined. Ill-conditioned means, vaguely, that small changes in the given numbers will produce large changes in the results.

The ill-conditioning may arise from three distinct sources. First, it may be the original problem itself. Thus a pencil balanced on its point (Fig. 7.8.1) presents us with a situation in which small changes in the initial position will result in large changes in the subsequent position in a short time. This is usually referred to as an *unstable system*.

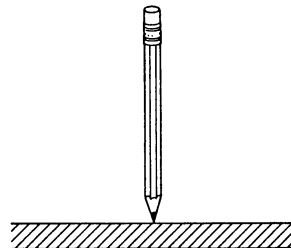


FIGURE 7.8.1

Second, the trouble may be not in the problem but in the representation used. For example, if we solve the differential equation

$$\frac{d^2 y}{dx^2} = y \quad y(0) = 1 \quad y(1) = -1$$

by using the hyperbolic functions as our basis for representation, then the solution is

$$y = \cosh x - \sinh x$$

and we will indeed see that small changes give rise to large differences later (Fig. 7.8.2). But if we choose the exponential functions (which are mathematically completely equivalent) as a basis, we get

$$y = e^{-x}$$

For large values of  $x$  there is no trouble with this representation.

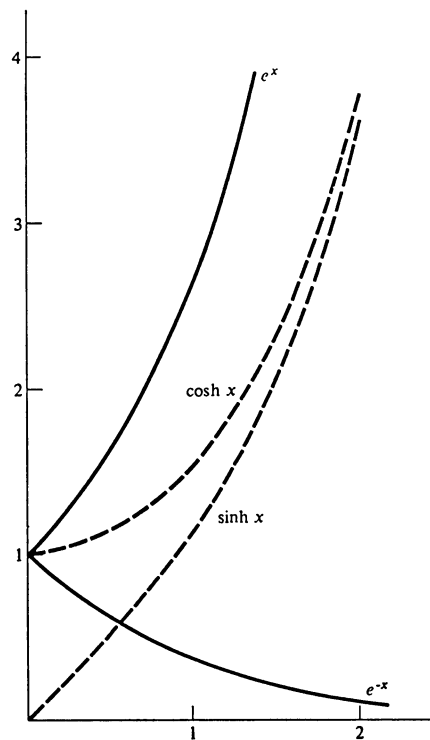


FIGURE 7.8.2

Third, the source of the ill-conditioning may be in the method of solution. Consider, for example, the following symmetric system of equations which appear to be reasonably scaled.

$$\begin{array}{rcl}
 3x + 2y + z & = & 3 + 3\varepsilon \\
 2x + 2\varepsilon y + 2\varepsilon z & = & 6\varepsilon \\
 x + 2\varepsilon y - \varepsilon z & = & 2\varepsilon
 \end{array} \quad \varepsilon \text{ small}$$

$x + \frac{2}{3}y + \frac{1}{3}z = 1 + \varepsilon$   
 $(-\frac{4}{3} + 2\varepsilon)y + (-\frac{2}{3} + 2\varepsilon)z = -2 + 4\varepsilon$   
 $(-\frac{2}{3} + 2\varepsilon)y + (-\frac{1}{3} - \varepsilon)z = -1 + \varepsilon$

For small  $\varepsilon$  these are ill-conditioned and are almost linearly dependent. But if we first eliminate  $x$  between the original second and third equations, we get

$$\varepsilon y - 2\varepsilon z = -\varepsilon$$

In floating-point arithmetic this is equivalent to

$$y - 2z = -1$$

and we can now solve, using the first two equations with this equation to get

$$y = z = 1$$

Put these in the first equation and we will have roundoff trouble, but in either of the other two we get accurately

$$x = \varepsilon$$

The fact that the system may be well conditioned and apparently reasonably scaled and that still the method of Gaussian elimination appears ill-conditioned is apparently not widely known nor its importance appreciated.

The conventional “scale by rows and then by columns, or vice versa,” may cause ill-conditioning. Given the matrix of coefficients

$$\begin{pmatrix} 3 & \frac{1}{\varepsilon} & \frac{1}{\varepsilon} \\ 1 & 2\varepsilon & \varepsilon \\ 2 & \varepsilon & \varepsilon \end{pmatrix}$$

where  $\varepsilon$  is small and due to roundoff effects we equate

$$1 \pm \varepsilon^2 = 1$$

Scale by rows first

$$\begin{pmatrix} 3\varepsilon & 1 & 1 \\ 1 & 2\varepsilon & \varepsilon \\ 2 & \varepsilon & \varepsilon \end{pmatrix}$$

pivot ↗

$$2 \begin{pmatrix} 0 & 1 & 1 \\ 0 & \frac{3\varepsilon}{2} & \frac{\varepsilon}{2} \\ 1 & \frac{\varepsilon}{2} & \frac{\varepsilon}{2} \end{pmatrix}$$

↖ pivot

$$2 \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & -\varepsilon \\ 1 & \frac{\varepsilon}{2} & \frac{\varepsilon}{2} \end{pmatrix}$$

and the back  
substitution is easy

Scale by columns first

$$\begin{pmatrix} 3 & 1 & 1 \\ 1 & 2\varepsilon^2 & \varepsilon^2 \\ 2 & \varepsilon^2 & \varepsilon^2 \end{pmatrix}$$

↖ pivot

$$\begin{pmatrix} 3 & 1 & 1 \\ 0 & -\frac{1}{3} & \frac{1}{3} \\ 0 & -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

and the rank is 2

**PROBLEMS**

- 7.8.1 Devise another example of apparently scaled systems which are ill-conditioned by straight Gaussian elimination but are in fact not ill-conditioned when solved properly.
- 7.8.2 Scale the “row and column scaling” example by the method of Sec. 7.6.

**7.9 THE RIGHT-HAND SIDES CAN CAUSE ILL-CONDITIONING**

It is widely believed that ill-conditioning depends *only* on the coefficients of the unknowns, but this is false as the following example shows. Take the same symmetric system as in Sec. 7.8 and change the right-hand sides to form the system

$$\begin{aligned} 3x + 2y + z &= 6 \\ 2x + 2\varepsilon y + 2\varepsilon z &= 2 + 4\varepsilon \\ x + 2\varepsilon y - \varepsilon z &= 1 + \varepsilon \end{aligned}$$



The solution is  $x = y = z = 1$ , and the system is indeed ill-conditioned because we are unable to solve the system so that the factor  $\varepsilon$  can be *divided out* as we did before *without* depending on the exact cancellation of the integer parts on the right-hand side (which, of course, will produce a large shift in the floating point addition) and hence depend on the rounded-off parts of the  $\varepsilon$  terms. As a result the answers are sensitive to relatively small changes in the coefficients of  $x$ .

### PROBLEMS

- 7.9.1 Explain “why” the above example works. Devise another example showing this effect.
- 7.9.2 It is often thought that it is the angle between the lines that makes ill-conditioning. For  $N = 2$  show that this statement is false.

### 7.10 A DISCUSSION OF GAUSSIAN ELIMINATION

Gaussian elimination is the main tool for solving simultaneous linear equations. In spite of all the faults and misconceptions about the method it is often reasonably effective. How can this be? As we observed earlier, many systems are naturally scaled by the source from which they come.

Consider pivoting next. Once the system is scaled, the value of the determinant is fixed, and the larger the early pivots are, the smaller the later ones must be. It is not correct that big pivots are good and small pivots are *automatically* bad; rather, large numbers become large *only* through multiplication by large numbers, while small numbers can be small because of multiplications by small numbers *or else* by heavy cancellation. Thus a large number is comparatively safe, while a small one is suspect. By picking the large numbers as pivots we tend to avoid the numbers that were formed by heavy cancellation.

But we may doubt the wisdom of the basic method. Given a system of equations and the general processes used in eliminating one variable, if we ask how to produce the maximum amount of linear dependence in the next stage, then we would be tempted to use one equation to eliminate from all the others so that at the next stage everywhere we look we would see this same one equation—that is, we would do something like pivoting. It is possible instead to eliminate a variable using adjacent equations thus avoiding the same equation in every derived equation. We can clearly see a mechanism that produces ill-conditioning in Gaussian elimination if we imagine a system of equations scaled and with the pivot in the

top equation, but also with the largest numbers in two other columns being in the top equation (Fig. 7.10.1). When we use the pivot equation, these largest two numbers tend to dominate all the derived equations, thus producing a linear (column) dependence. Indeed, looking at some of the earlier examples, we can see this is the mechanism behind why the examples “worked” (i.e., failed).

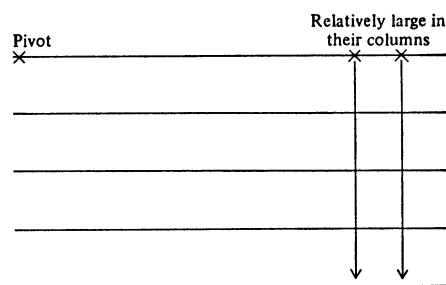


FIGURE 7.10.1

The user should be wary of the use of bounds as a measure of the quality of the results and especially as the basis for a comparison of two methods. Bounds tell only part of the story. The answer may be very much better than the careful bounds indicate. Backward analysis which pushes the errors committed back onto the original coefficients and states that the numbers obtained are a solution to a nearby system of equations is of little use *unless* the zero coefficients are kept as zero in the nearby system. (We should use the floating-point concept of relative error and not the fixed-point idea of additive error—a point that is rarely noticed in backward analysis.)

In many places in the literature the condition of a system of equations is given in terms of the singular values of the coefficient matrix. The singular values of a matrix  $A$  are the eigenvalues of  $A^T A$  (see Chap. 45). Of course if  $A$  is symmetric, then the singular values are just the squares of the eigenvalues. But Sec. 7.9 shows that this cannot be the whole story.

If the coefficient matrix is symmetric, then it can be written as the sum of its principal idempotents (projections), each multiplied by an eigenvalue. The accuracy of the solution can then be related to this expression, as is well summarized by Richard Rosanoff<sup>1</sup>:

<sup>1</sup> Rosanoff and Nishimoto, Zeus on the Leus, Space Division, North American Rockwell Corp.

Idempotents summed are the  
 Array baby,  
 When truncated they create  
 Dismay baby,  
 You cannot invert them  
 Any way baby,  
 Gauss can't give you anything  
 But love.

### 7.11 MATRIX INVERSION 1

The problem of finding the inverse of a square matrix  $A = (a_{i,j})$  is very common. Since matrices are not in general commutative, there is the idea of a left inverse and a right inverse

$$A_L^{-1}A = I$$

$$AA_R^{-1} = I$$

It is an important mathematical result that these two inverses are the same. The proof follows easily from the associativity.

$$A_L^{-1}AA_R^{-1} = \begin{cases} (A_L^{-1})(AA_R^{-1}) = (A_L^{-1}I) = (A_L^{-1}) \\ (A_L^{-1}A)(A_R^{-1}) = (IA_R^{-1}) = (A_R^{-1}) \end{cases}$$

However, this may be very far from true for some computed matrices. An inverse may be close when multiplied on one side (in the sense that the off-diagonal elements are small) but very far when multiplied on the other side. For example, let

$$A = \begin{pmatrix} 1.00 & 1.00 \\ 1.00 & 0.99 \end{pmatrix} \quad C = \begin{pmatrix} -89 & 100 \\ 90 & -100 \end{pmatrix}$$

then

$$AC = \begin{pmatrix} 1.0 & 0.0 \\ 0.1 & 1.0 \end{pmatrix} \quad CA = \begin{pmatrix} 11 & 10 \\ -10 & -9 \end{pmatrix}$$

Given the coefficients of a square matrix  $A = (a_{i,j})$ , if we were to solve the systems with the right-hand sides

first     1, 0, 0, ..., 0  
 second    0, 1, 0, ..., 0  
 third     0, 0, 1, ..., 0  
 etc.

and arrange the solutions in a matrix form with the solutions as columns, then we would have the inverse. To see why this is so consider what you would do if you had these special fundamental solutions and were asked to solve the same system with the right-hand sides  $b_1, b_2, b_3, \dots, b_N$ . You would multiply these fundamental solutions in turn by  $b_1, b_2, \dots, b_N$  and add the results.

Thus, given

$$Ax = b$$

we have

$$x = A^{-1}b$$

which is what you did to get the answer—you multiplied the  $(b_i)$  by a matrix.

### PROBLEMS

7.11.1 Find an example when  $AA^{-1}$  and  $A^{-1}A$  are very different.

7.11.2 Find the inverse of

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

7.11.3 Find the inverse of

$$\begin{pmatrix} 0 & 2 & 3 \\ 3 & 0 & 2 \\ 3 & 2 & 0 \end{pmatrix}$$

## 7.12 MATRIX INVERSION 2

A completely equivalent method of finding the inverse is to write the  $2N \times N$  matrix

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N} & 1 & 0 & \cdots & 0 \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N} & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{N,1} & a_{N,2} & \cdots & a_{N,N} & 0 & 0 & \cdots & 1 \end{pmatrix}$$

and to use the steps of the Gauss-Jordan elimination to transform the first  $N$  columns into the identity matrix. We will find that the last  $N$  columns are the inverse of the first  $N$  columns. To see this we must first recognize that each step we used can be written as multiplication on the left by some matrix. When we

are done, we can, by the associative law of multiplication, gather all the matrices we used into one single matrix, say  $C$ . We have

$$CA = I$$

hence

$$C = A^{-1}$$

But the matrix  $C$  also operated on the last  $N$  columns (the identity matrix), so when we are done, we have  $A^{-1}$  in the last  $N$  columns.

**EXAMPLE 7.1** Given the matrix

$$\begin{pmatrix} 1 & 1 & 4 \\ -1 & 0 & 1 \\ 1 & -1 & -5 \end{pmatrix}$$

form

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 1 & 0 \\ 1 & -1 & -5 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & 5 & 1 & 1 & 0 \\ 0 & -2 & -9 & -1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & 5 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & -3 & -8 & -4 \\ 0 & 1 & 0 & -4 & -9 & -5 \\ 0 & 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & -4 & -9 & -5 \\ 0 & 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

The inverse is

$$\begin{pmatrix} 1 & 1 & 1 \\ -4 & -9 & -5 \\ 1 & 2 & 1 \end{pmatrix} \quad ////$$

## PROBLEMS

Using the method of Sec. 7.12, find the inverse of the following matrices and check the results.

7.12.1  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$7.12.2 \quad \begin{pmatrix} 1 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$7.12.3 \quad \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

$$7.12.4 \quad \begin{pmatrix} 2 & 3 & 5 \\ 7 & 11 & -13 \\ 17 & -19 & 23 \end{pmatrix}$$

## REFERENCES

Probably the two best references to this highly developed field are Forsythe [11] and Noble [46].

# 8

## \*RANDOM NUMBERS

### 8.1 WHY RANDOM NUMBERS?

Randomness is traditionally looked upon as something to be avoided if possible. The idea that randomness may be used constructively strikes many people as strange. However, many parts of modern computing practice use random numbers, and we shall give a few examples in this chapter.

In spite of formal axioms, probability is an intuitive idea, just as in Euclidean geometry the ideas of point, line, and plane are intuitive. Randomness is closely connected with probability. As an example of how the modern, complex idea of a random process can arise, consider the simple act of “tossing a well-balanced coin.” If we record a 1 for a head and a 0 for a tail, we have a simple random variable. If next we consider not a single toss but a sequence of tosses, then we have a sequence of 0s and 1s, which we may regard as a binary number. Thus we are led by easy stages into gradually considering an indefinite number of tosses and then to regarding the resulting sequence of 0s and 1s as the binary representation of a number between 0 and 1. As a result we have for each of the nondenumerable numbers in  $0 \leq x < 1$  a corresponding sequence of tosses of a

simple coin. This is the idea of a random, or stochastic, process; corresponding to some index  $\lambda$  we have a family of functions

$$f_{\lambda}(t)$$

where the variable  $t$  is often regarded as time, either continuous or at discrete intervals.

With our finite machine we cannot hope to actually carry out the mathematician's random process; instead we will approximate it by some finite sequence of finite numbers. Furthermore, as we shall show, in most cases we will not deal with what is intuitively random numbers, but rather we will generate a perfectly definite sequence that is completely predictable. This approach violates the view of random that is usually equated with nonpredictable. However, in practice we do not care if the numbers are theoretically predictable. We care only that *with regard to their use* they are random.

The idea of a definite random process is almost a contradiction, and so we must approach the matter slowly. First we examine a couple of very simple examples to see how random numbers might be used in many applications and why we are not alarmed at their actual structure in these cases. Evidently we would not use a predictable sequence of random numbers in such applications as lottery selections and other processes that interact directly with humans; we use them only with machines and then *only* with care.

After looking at the simple uses, we will look both at possible sources of uniformly distributed random numbers and at why we prefer a predictable source. Only then will we look at how to generate them and prove that they have certain properties. Since it is a difficult topic, we will give a few moments' consideration to the complex matter of testing random numbers.

We then look at various distributions other than the flat one and at ways of using various tricks from statistics to aid in saving machine time. As is so often the case, what we cover in one short chapter can be, and has been, expanded into a whole book.

## 8.2 SOME USES OF RANDOM NUMBERS

The most obvious use of random numbers is in the simulation of random processes such as nuclear disintegrations, people coming at random for a particular service, telephone traffic studies, and making random choices in some decision process, say playing a game.

Less obvious are the so-called "Monte Carlo" applications in which a definite process is replaced by a random process that arrives at the same result.



The most famous example is probably the Buffon<sup>1</sup> needle. In 1773 Buffon observed that if a needle of length  $L \leq 1$  were tossed at random onto a horizontal surface ruled with equally spaced lines, say at unit spacing, then the probability of the needle crossing a line is

$$\text{Probability} = \frac{2L}{\pi}$$

He reasoned, therefore, that he could experimentally determine the value of  $\pi$  by making repeated trials.

Let us derive this result (see Fig. 8.2.1). The position of the needle's center is within  $1/2$  unit of some line, and the words "tossed at random" mean that the probability is uniform in the interval  $0 \leq x \leq \frac{1}{2}$ . The angle  $\theta$  that the needle

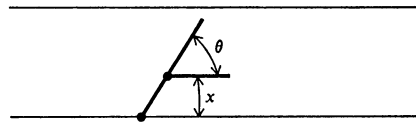


FIGURE 8.2.1

makes with the direction of the lines varies in the interval  $0 \leq \theta < \pi$ , and again the words "tossed at random" are taken to mean that the probability distribution is uniform in the interval. A crossing will occur if and only if

$$x \leq \frac{L}{2} \sin \theta$$

From Fig. 8.2.2 we see that this probability is the ratio of the shaded area to the area of the whole rectangle

$$\text{Probability} = \frac{\int_0^{\pi} (L/2) \sin \theta \, d\theta}{\int_0^{\pi} \frac{1}{2} \, d\theta} = \frac{2L}{\pi}$$

If the value of  $\pi$  were inaccurately known, then this would be a good method for finding that it is around 3. With some care in drawing the lines, measuring the needle and the crossings, and tossing at random, we could get perhaps 3.1 with some reliability; but to get 3.14 would tax our abilities and patience.

Many generalizations of this famous needle problem have been used. One,

<sup>1</sup> Georges Louis Leclerc, Comte de Buffon (1707–1788).

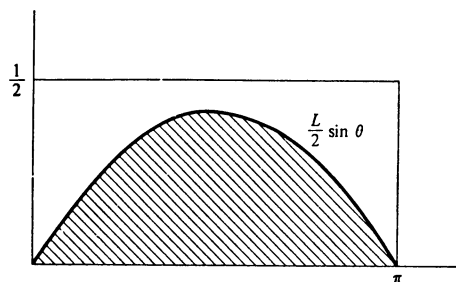


FIGURE 8.2.2

for example, estimates the amount of “grain boundary” in microphotographs of magnetic materials.

A similar Monte Carlo method,<sup>1</sup> this time estimating  $e$ , is based on the observation that if  $2k$  numbers  $x_i$  are drawn in sequence from a random equilikely source, then the probability that they are all in ascending order  $x_1, x_2, x_3, \dots, x_{2k}$  is

$$\frac{1}{(2k)!}$$

This probability is easily found from the observation that the  $2k$  numbers  $x_i$  can have  $(2k)!$  possible orderings and that only one ordering will be in ascending order. The probability that a sequence of trials yielding an increasing sequence of  $x_i$ 's will fail on the odd trial  $2k + 1$  is the difference

$$\frac{1}{(2k)!} - \frac{1}{(2k+1)!}$$

Thus the total probability that a sequence of drawings of random numbers from an equilikely source will produce a rising sequence that ends with an even number of numbers is

$$\sum_{k=1}^{\infty} \left[ \frac{1}{(2k)!} - \frac{1}{(2k+1)!} \right] = \sum_{k=0}^{\infty} \left[ \frac{1}{(2k)!} - \frac{1}{(2k+1)!} \right] = \frac{1}{e}$$

An experiment using 252 runs gave  $1/e = 0.381$  (a 3.5 percent error).

In both of these examples the experiment is a test more of the source of the random numbers used than of the result. Indeed, in the second case it provided a nice, simple check on the derivation of the formula—the test would have caught the elusive factor of 2, the slip in the minus sign, and many other typical human errors in mathematics. It is a common research practice to “experimentally

<sup>1</sup> Due to Roger Pinkham, (1929—).

verify some result" by some simple numerical tests before building further on the result, and Monte Carlo techniques are one more tool for doing this.

Another use of random numbers, which are most often random decimal digits (base 10) or random bits (base 2), is in game playing on a machine. When things are equally favorable, then often it is desirable to make a random choice—this, for example, will tend to defeat the opponent who tries to play the same opening repeatedly until he finds a weakness in the formulas used by the machine.

The uncritical use of randomness to solve complicated problems should be tempered with a dose of reality. It may well be true that by using random trials one can expect to find the solution, but the size of the task may be like the famous case of the monkeys and the typewriters. In this story a number of monkeys are set down at typewriters, and we wait until they type out in sequence all the books in the British Museum in the card-catalog order. There is a finite chance that the first letter will be typed correctly, and indeed, in the infinitude of time this will happen infinitely often. Among this infinite number of cases there is a positive probability that the second correct letter will be typed next, and among all these infinitely many cases of two letters . . . Often grandiose proposals to use a random search are this situation in some disguise! A random search should not be started unless some careful estimates of the running time are made—otherwise you are apt to get a very large computing bill and waste a lot of time in the preparation of the program with little hope of getting a result.

## PROBLEMS

8.2.1 Using the integral  $-\int_0^1 e^{-x} \ln x \, dx = \gamma$ , set  $e^{-x} = t$  and estimate  $\gamma$ , using 10 random points.

## 8.3 SOURCES OF RANDOM NUMBERS

The first source of random numbers likely to come to mind is some natural, physical phenomena such as cosmic rays, nuclear disintegrations, or perhaps sampling at random an alternating frequency. Such devices have been built in the past, and they do have their uses. Their main defect is that if they are used and something interesting happens in a computation, then it is practically impossible to repeat the run to find out how the effect came about. This foolishness is repeatedly done; a clever programmer may realize that by consulting the clock time at widely separated intervals and using the last digit of the time, he has what is probably a source of random digits, but he will fail to realize the consequences for the research because he thereby removes the repeatability of the computation.

Tables of random digits have been published, the most famous being the Rand Table of One Million Random Digits and One Hundred Thousand Normal Deviates.<sup>1</sup> In principle such tables could be stored on a tape and read into a machine as needed, but this is rarely done because of the slowness of the input and because in large problems with random numbers 1 million digits is hardly enough. It is possible to use a table to create a new table, for example by adding adjacent numbers and dropping the carry, but this does not seem to be popular, although this device was used by Rand to make their numbers more random.

The third source is mathematical formulas. Many sources have been proposed, such as the old "in a  $k$ -digit machine, square the last random number and keep the middle  $k$  digits of the product as the new random number," but few are kept in use for long. The ones most often used are still multiplicative congruence generators of the form

$$x_{n+1} \equiv x_n \rho \pmod{2^k}$$

where we take the last  $k$  digits of the fixed-point product (on a  $k$ -digit binary machine). In words, " $x_{n+1}$  is congruent to  $x_n$  modulo  $2^k$ " means the difference between  $x_{n+1}$  and  $x_n$  is divisible by  $2^k$ .

As an illustration of this kind of a random-number generator consider using 000 001 as  $x_0$  and 100 101 as the multiplier on a six-bit machine. We get Table 8.3 of products. As we can easily see, all the numbers are of the form  $4k + 1$ .

<sup>1</sup> Free Press, Chicago, 1955.

Table 8.3

| Binary number | Decimal equivalent | Drop last two digits |
|---------------|--------------------|----------------------|
| 000 001       | 1                  | 0                    |
| 100 101       | 37                 | 9                    |
| 011 001       | 25                 | 6                    |
| 011 101       | 29                 | 7                    |
| 110 001       | 49                 | 12                   |
| 010 101       | 21                 | 5                    |
| 001 001       | 9                  | 2                    |
| 001 101       | 13                 | 3                    |
| 100 001       | 33                 | 8                    |
| 000 101       | 5                  | 1                    |
| 111 001       | 57                 | 14                   |
| 111 101       | 61                 | 15                   |
| 010 001       | 17                 | 4                    |
| 110 101       | 53                 | 13                   |
| 101 001       | 41                 | 10                   |
| 101 101       | 45                 | 11                   |
| 000 001       | 1                  | 0                    |

The last binary digit is hardly random, nor is the next to the last, but dropping both of them, we have the decimal representations of the numbers 0, 1, ..., 15 in what looks like a random order. However, each number occurs only once in the cycle of 16 numbers, and that is definitely suspicious.

As we will prove in Sec. 8.4, using any  $(k \geq 3)$ -bit machine will have the same effects. All the numbers, once the last two digits are dropped, are represented exactly once, but otherwise the numbers seem to be in a chaotic order. The periodicity is for the typical 35-bit machine

$$2^{k-2} = 2^{33} \approx 8.59 \times 10^9$$

and seems to be adequately long for most simple applications.

### PROBLEMS

- 8.3.1 Make a random-number generator using a 5-bit word length with  $\rho = 13$  and  $x_0 = 1$ .
- 8.3.2 Find the period of a 6-bit random-number generator using  $\rho = 33$  and  $x_0 = 1$ .
- 8.3.3 Using the generator in the text, remove the last 3 bits and study the probability of one number being followed by another, i.e., the correlation of adjacent numbers.
- 8.3.4 Repeat the random-number generator in the text except start with  $x_0 = 3$ . Compare the two runs.

## 8.4 THE RANDOM-NUMBER GENERATOR

We now study random-number generators of the form

$$x_n \equiv \rho x_{n-1} \pmod{2^k}$$

where  $\text{mod } 2^k$  means “keep the remainder after dividing the product by  $2^k$ ,” and this, in a binary machine, is simply the last  $k$  digits of the product, the lower part of the accumulator. How do we pick the multiplier  $\rho$  and the starting value, and what can we expect for the resulting sequence of random numbers?

If 2 divides  $x_0$ , then all the products will have at least one 0 at the end, and we are wasting one bit of the machine capacity. Hence we take  $x_0$  as an odd number.

Similarly, if  $\rho$  is even, then the products will gradually accumulate 0s on the right-hand end until

$$x_{k+1} = \rho^{k+1} x_0 = 0 \pmod{2^k}$$

and all numbers past this point will be zero.

Therefore  $\rho$  is taken as an odd number.

Now all odd numbers can be written in one of the forms

$$8t + 3 \quad 8t + 1 \quad 8t - 1 \quad 8t - 3$$

for some integer  $t$ . How shall we pick  $t$ , and which of the four forms should we use?

**Theorem 8.4.1** If  $\rho = 8t \pm 3$ , then the period of the sequence is  $2^{k-2}$ .

**PROOF** If the period is to be  $2^{k-2}$ , then the theorem states that

$$\rho^{2^{k-3}} \not\equiv 1 \pmod{2^k}$$

$$\rho^{2^{k-2}} \equiv 1 \pmod{2^k}$$

which is equivalent to saying  $2^k$  divides  $(\rho^{2^{k-2}} - 1)$  but does not divide  $(\rho^{2^{k-3}} - 1)$ .

We begin by observing that using

$$a^2 - 1 = (a + 1)(a - 1)$$

repeatedly we can write

$$(\rho^{2^{k-2}} - 1) \equiv (\rho^{2^{k-3}} + 1)(\rho^{2^{k-4}} + 1) \cdots (\rho + 1)(\rho - 1)$$

For each factor ( $i \geq 1$ )

$$\begin{aligned} \rho^{2^i} + 1 &= 1 + (3 \pm 8t)^{2^i} \\ &= 1 + 3^{2^i} + \sum_{k=1}^{2^i} (\pm 1)^k C(2^i, k) (8t)^k 3^{2^i-k} \end{aligned}$$

But

$$\begin{aligned} 1 + 3^{2^i} &= 1 + (4 - 1)^{2^i} = 1 + (1 - 4)^{2^i} \\ &= 1 + 1 + \sum_{k=1}^{2^i} (-4)^k C(2^i, k) \end{aligned}$$

so that

$$\rho^{2^i} + 1 = 2 + \sum_{k=1}^{2^i} C(2^i, k) [(\pm 1)^k 3^{2^i-k} t^k 8^k + (-4)^k]$$

and 2 divides  $\rho^{2^i} + 1$ , but 4 does not.

Returning to the factored form, we use this in each of the first  $2^{k-3}$  terms and conclude that

$$2^{k-3} \text{ divides } (\rho^{2^{k-3}} + 1)(\rho^{2^{k-4}} + 1) \cdots (\rho^2 + 1)$$

but  $2^{k-2}$  does not divide it.

We have left the two factors

$$\begin{aligned}(\rho + 1)(\rho - 1) &= \rho^2 - 1 = (8t \pm 3)^2 - 1 \\ &= 64t^2 \pm 48t + 9 - 1 \\ &= 8(8t^2 \pm 6t + 1)\end{aligned}$$

and  $2^3$  divides it, but  $2^4$  does not. Thus we have

$$2^{k-3} \cdot 2^3 = 2^k$$

divides the original

$$\rho^{2^{k-2}} - 1$$

but no higher power does.

Rearranged, we have

$$\begin{aligned}\rho^{2^{k-3}} - 1 &\not\equiv 0 \pmod{2^k} \\ \rho^{2^{k-2}} - 1 &\equiv 0 \pmod{2^k} \quad \text{////}\end{aligned}$$

**Theorem 8.4.2.** If  $\rho = 8t - 3$ , then the sequence  $x_0, x_1, \dots, x_{2^{k-2}-1}$  generated by the formula is some permutation of

$$\begin{aligned}1, 5, 9, \dots, 2^k - 3 &\quad \text{if } x_0 \equiv 1 \pmod{4} \\ 3, 7, 11, \dots, 2^k - 1 &\quad \text{if } x_0 \equiv 3 \pmod{4}\end{aligned}$$

**PROOF** Consider the numbers

$$\rho^n \quad n = 0, 1, \dots, 2^{k-2} - 1$$

The difference between consecutive terms

$$\begin{aligned}x_0 \rho^{n+1} - x_0 \rho^n &= x_0 \rho^n (\rho - 1) \\ &= x_0 \rho^n (8t - 4)\end{aligned}$$

is divisible by 4. We know that the period is  $2^{k-2}$ ; hence we have  $2^{k-2}$  distinct numbers whose differences are all divisible by 4, and the theorem follows from this. ////

How shall we pick the value of  $t$ ? For example, if  $t = 1$ , then  $\rho = 5$  and whenever a small  $x_i$  occurs then for some time the succeeding  $x_i$  will increase—definite structure in the supposedly random sequence. Similarly for  $\rho$  near 1.0. . . . It is customary to pick a  $t$  so that  $\rho$  is near 1/2, say  $\rho = 0.10$ . . . .

If in the sequence  $x_i$  of random numbers we drop the last two bits, then we will have all possible  $(2^{k-2})$ -bit numbers in some order. In practice it is customary to avoid more than the last two bits but to recognize that dropping many more does not help very much.

## PROBLEMS

- 8.4.1 Examine the periods for  $p = 8t \pm 1$  and  $p = 8t \pm 3$ .  
 8.4.2 Discuss the cycles that all the  $2^k$  numbers fall into.  
 8.4.3 In picking a multiplier note that *both* ends need to be considered. A multiplier ending in  $\dots 01$  will reproduce exactly the last two digits, and one in  $\dots 11$  will cause the last digit to alternate. Extend this analysis to numbers ending in  $\dots 001$  and others.

## 8.5 TESTING A RANDOM-NUMBER GENERATOR

Now that we know what we are going to use as a source of random numbers it is necessary to consider how well it meets our intuitive notions of what a random sequence of numbers should be.

In the first place we have implied that the numbers should be equilikely, that is, we were talking about a uniform distribution in an interval, which is conventionally taken as  $0 \leq x < 1$  by putting a binary point in front of the first digit. It is easy to see from Theorem 8.4.2 that until we get close to the granularity of the number system, we have an *exactly* uniform distribution in the full period. This bothers us a bit since it is too uniform. We are sampling without replacing until we exhaust the whole set of numbers we are going to use, and *random* intuitively means that there is a chance, small though it may be, that the same number can come up twice in succession. Indeed there is a reasonable probability that in the long sequence there would be one such case. But this defect is not serious for most applications, and if we drop more digits on the right, we can get sequences in which such numbers occur a fixed number of times. Again this is too uniform and is not compatible with our intuitive feelings.

Taking the numbers two at a time, we can study the pattern of number pairs, which we again expect to be uniform. It is possible to show theoretically that the number pairs are not bad, but carried far enough by studying triples, etc., we will be in trouble. In practice the effect is seldom serious.

There are almost an infinite number of tests one can apply to a random-number sequence, run tests, poker-hand draws, power spectral tests, etc., and a large computer could be run full-time testing various generators. Randomness is essentially a negative property and cannot be proved in a mathematical sense by any finite amount of testing of a finite length of data. Every finite string of random numbers will be found to have some peculiar property (even if it is only that it does not have a peculiar property!). The topic of testing fascinates some people, and if they wish, they can follow the topic further in the references at the end of the chapter.



Experience, which is not an infallible guide, indicates that with care congruential generators give useful results, but carelessness regularly trips up the unwary. The use of an additional operation of adding a constant to the product before taking the last  $k$  digits has not been proved worth the time and trouble.

In the final analysis, randomness, like beauty, is in the eye of the beholder, and how the numbers are to be used determines their randomness.

## 8.6 OTHER DISTRIBUTIONS

We now have a source of random numbers with a flat distribution. In many applications other distributions are needed. In principle the following device works. We equate the cumulative distribution  $F(y)$  of the distribution  $f(y)$  that we want to the cumulative distribution  $x$  of the flat distribution and solve for the new one

$$\int_0^x 1 \, dx = x = \int_0^y f(y) \, dy = F(y)$$

Applying the inverse operation  $F^{-1}$ , we have

$$F^{-1}\{F(y)\} = y = F^{-1}(x)$$

For example, suppose we want the exponential distribution  $e^{-y}$ . Then

$$\begin{aligned} f(y) &= e^{-y} \\ F(y) &= \int_0^y e^{-y} \, dy = 1 - e^{-y} \\ e^{-y} &= 1 - x \\ y &= -\ln(1 - x) \end{aligned}$$

and using the sequence of random numbers  $x_i$  from a flat distribution, we get

$$y_i = -\ln(1 - x_i)$$

or equally good,

$$y_i = -\ln x_i$$

This has been used many times with excellent success.

If this method of inverting the cumulative function is used on the normal distribution

$$\frac{e^{-y^2}}{\sqrt{2\pi}}$$

then the inverse function has to be approximated somehow—perhaps with the aid of auxiliary tables.

An alternate method for getting a normal distribution is to appeal to the central-limit theorem and simply add several random numbers from a flat (or any other) distribution. Since the numbers in the source are independent and have a variance,

$$\sigma^2 = \int_0^1 (x - \frac{1}{2})^2 dx = \frac{(x - \frac{1}{2})^3}{3} \Big|_0^1 = \frac{1}{12}$$

The sum of 12 of them will have a variance of 1. Thus the common rule is to “add 12 numbers from our generator and subtract the constant 6 from the sum to get a normal distribution with mean zero and unit variance.”

How good is this distribution? Clearly it is exactly zero outside of the interval  $\pm 6\sigma$ . The probability of a value being outside is approximately  $2 \times 10^{-9}$  and hardly has a serious effect.

This brings up a philosophical point. Do we really want genuine random numbers, or do we want a set of homogenized, guaranteed, and certified numbers whose effect is random but at the same time we do not run the risk of the fluctuations of a truly random source? When we look at how we are using the numbers, we usually find that we want to get the security of a large sample by taking as small a sample as we can. We do not want to run the risk of wild fluctuations. Thus our too-flat random generator and the normal distribution which eliminates the tails far out are closer to what we often want than the real thing is!

Many other distributions can be found by various methods described in the two references at the end of the chapter.

## 8.7 RANDOM MANTISSAS

One of the many uses of random numbers is in the simulation of random computations, or as a source of random data to test a routine. As we showed in Sec. 2.8, the distribution of mantissas is not flat but rather has the reciprocal distribution

$$p(x) = \frac{1}{x \ln b}$$

This distribution also applies to random physical data or random data for testing a mathematical library routine for average running time, etc. Thus we need a source of random mantissas.

One of the results proved in Sec. 2.8 is that a continued product of numbers from any reasonable distribution has a mantissa that rapidly approaches the reciprocal distribution. We gave a short table showing the rapidity of approach for numbers from our flat distribution.

A specific algorithm based on this observation uses the  $x_i$  from the random-number generator and forms

$$Y_n = Y_{n-1}x_n \quad (\text{shifted})$$

where *shifted* means that after the multiplication the possible leading zeros are shifted off to form a legitimate floating-point mantissa.

How well does this method work? Experimental verification is given by 8,192 trials. Counting the number of mantissas falling in each of  $N$  categories, we get the results shown in Table 8.7. The last two columns give the number of sign changes in the residuals between the observed and theoretical distributions. The expected number of sign changes might be thought to be  $(N - 1)/2$ , but since for  $N = 2$  it is clear that one sign change must occur (because the mean of the residuals is zero), we have used  $N/2$  as the expected number. A comparison of these two columns shows that there are no systematic errors in some regions. The chi-square test in the second column shows that the size of the deviations is approximately right, neither too small nor too large. Thus the generator appears to be safe to use.

It is interesting to note that probably this generator has a significantly longer period than the period of the source  $x_i$ .

## PROBLEMS

8.7.1 Apply this generator to the 6-bit random numbers of Sec. 8.3.

Table 8.7 DISTRIBUTION OF 8,192 RANDOM MANTISSAS

| $N$ | $\chi^2$ | Degrees of freedom | Residuals    |          |
|-----|----------|--------------------|--------------|----------|
|     |          |                    | Sign changes | Expected |
| 64  | 61.392   | 63                 | 30           | 32       |
| 32  | 22.804   | 31                 | 14           | 16       |
| 16  | 11.150   | 15                 | 8            | 8        |
| 8   | 7.724    | 7                  | 5            | 4        |
| 4   | 3.261    | 3                  | 2            | 2        |
| 2   | 1.467    | 1                  | 1            | 1        |

## 8.8 SWINDLES

Problems using random numbers are often both tricky and inclined to be expensive in machine time; therefore it is wise to have a competent practicing statistician handy to discuss the computation before you begin.

A number of methods have been devised to make Monte Carlo computations and other computations using random numbers much more efficient, sometimes by factors of 1,000 or more. These are known by the colorful name of *swindles* to go along with the name *Monte Carlo*. One of the methods is called “antithetic variables,” meaning that when one case occurs, then it is negatively correlated in a known way with another case. For example, in the Buffon needle case a cross is used, instead of a needle, so that if one arm is likely to cross a line, then the other is not.

Many other statisticians’ tricks can be used, such as stratified sampling. Again, perhaps an analytical solution can be found for the bulk of the cases and the random sampling used only for the small difference. There are no systematic methods in the field as yet; the low cunning of trickery seems to be necessary for success, and this art lies outside the scope of a general textbook.

## 8.9 NOISE SIMULATION

Probably the greatest use of random numbers is in the simulation of noise. Rarely (beyond simple roundoff) does the noise resemble the flat distribution of our generator. For *independent* samples the transformation to other distributions is easy in principle, and since it does not need to be extremely accurate in most cases, it is fairly easy in practice. But most simulations of noise require that the samples be dependent from step to step and that they have various internal correlations. This topic is much too difficult to take up here, and often the proper choices depend on knowledge of the field of application. The proposer of a problem may not realize that he does not have independent samples of noise at the step size to be used in the computation, and care is necessary to avoid elegantly solving the wrong problem.

## REFERENCES

The three best references are Jansson [22], Knuth [28], and Shreider [51], with Knuth being the more complete and latest one.

# 9

## THE DIFFERENCE CALCULUS

### 9.1 INTRODUCTION

We have examined some algorithms for solving a few of the more common problems of numerical computations. Before examining the next major group of problems, we need to develop a few mathematical tools. The main tool needed is the *finite difference calculus*, which corresponds to the usual infinitesimal calculus *except* that we do not go to the limit but instead stop at a definite, finite step size. One reason for the importance of the finite difference calculus is that most of the time we have data (samples) of our functions at a sequence of equally spaced points  $x_k$  ( $k = 0, 1, \dots, N$ ),

$$f(a), f(a + h), f(a + 2h), \dots, f(a + Nh)$$

An example is the sequence of partial sums of a series

$$s(n) = \sum_{k=0}^n a_k \quad n = 0, 1, 2, \dots, N$$

A second reason is that when the difference is divided by the step size  $h$ , it provides an estimate for the derivative (which is a limit process and cannot be carried out by a finite machine).

If there is difficulty in understanding the finite difference calculus, then it is probable that the corresponding part of the infinitesimal calculus is not well understood and should be reviewed.

Because we are dealing with a finite number of steps and a finite number of operations, we do not get into the area of existence theorems or other difficult questions in mathematics; it is usually clear that we can carry out the proposed operations and that a number will result. Its relation to what is wanted may be obscure, but that is not the point at the moment. Because of the finite nature of our computations we can operate formally without regard to the usual questions of rigor.

It is worth recalling that differential equations, for example, are usually derived from some physical situation by first finding a finite approximation and then taking the limit. If the reverse step cannot be taken (when small steps are used), then there is some doubt about the validity of the differential equation. Thus it is reasonable to assume that the finite difference approximations we make are likely to be fairly accurate. Later we will examine the errors of such approximations. In this chapter and the next four, we will be dealing with finite, equally spaced data in a formal manner.

If the data are at the points

$$x = a, a + h, a + 2h, \dots$$

we can reduce the situation to a standard form by the substitution

$$t_k = \frac{x_k - a}{h} = k \quad k = 0, 1, 2, \dots$$

Much of the time we will assume that this has been done. It is often convenient to write

$$f(0) = f_0$$

$$f(1) = f_1$$

$$\dots\dots\dots$$

$$f(k) = f_k$$

In using these results we will have to either reduce the problem to the standard form or transform our formulas to the spacing of the original problem. Often it is a matter of indifference which we do, though once in a while it does make a significant difference in the amount of arithmetic that must be done. We can, as in analytic geometry, regard the transformation as an "alias" which keeps the

points fixed and moves the coordinate system, thus giving the points new names; or we can regard the transformation as an “alibi” which keeps the coordinate system fixed and regard the problem as having been moved. For example, the transformation

$$\begin{aligned}x' &= x - a \\y' &= y - b\end{aligned}$$

can be regarded as either shifting the coordinates of the circle

$$(x - a)^2 + (y - b)^2 = r^2$$

so that it is now labeled

$$x'^2 + y'^2 = r^2$$

or else as moving the circle to the origin. Whichever way we view it, we must keep that view to avoid unnecessary confusion.

## PROBLEMS

9.1.1 If  $x$  takes the values  $11, 9, 7, \dots, -11$ , find the transformation that reduces this to the standard form  $0, 1, \dots, 11$ . *Ans.*  $t = (11 - x)/2$

9.1.2 Reduce  $3.0, 3.5, 4.0, 4.5, \dots, 10$  to standard form. *Ans.*  $t = 2(x - 3)$

9.1.3 Reduce to standard form:  $x = a, a - h, a - 2h, \dots, a - (n - 1)h$ .

$$\text{Ans. } t = \frac{a - x}{h}$$

9.1.4 Transform Simpson's formula

$$\int_{-1}^1 f(x) dx = \frac{1}{3}[f(-1) + 4f(0) + f(1)]$$

to the interval  $a \leq x \leq b$ .

$$\text{Ans. } \int_a^b f(x) dx = \frac{b - a}{6} \left[ f(a) + 4f\left(\frac{a + b}{2}\right) + f(b) \right]$$

9.1.5 Transform the interval  $a \leq x \leq b$  to the interval  $c \leq t \leq d$ , assuming  $a, b, c, d$  finite.

$$\text{Ans. } t = \frac{(c - d)x + ad - bc}{a - b}$$

9.1.6 Transform the interval  $0 \leq x \leq \infty$  to the interval  $-1 \leq t \leq 1$ .

$$\text{Ans. } t = \frac{x - 1}{x + 1} \quad \text{or} \quad x = \frac{1 + t}{1 - t}$$

## 9.2 THE DIFFERENCE OPERATOR

The basic operation of the finite difference calculus is the difference operator  $\Delta$  defined by

$$\Delta f(x) \equiv f(x+h) - f(x)$$

This operation is familiar from the calculus where it is used in the process of finding a derivative.

We can imagine the operator  $\Delta$  as separated from the function on which it operates, just as we imagine the derivative operator  $d/dx$  operating on a function  $f(x)$  or the integration operator  $\int \dots dx$  acting on a function  $f(x)$ .

The difference operator is linear, as are differentiation and integration. By this we mean that if  $a$  and  $b$  are constants, then

$$\Delta (af(x) + bg(x)) = a \Delta f(x) + b \Delta g(x)$$

This property of being linear is very important and makes the operator  $\Delta$  very convenient to use. Finding zeros of a function is not a linear operation.

As an example of the linearity and application of the  $\Delta$  operator, consider applying it to the quadratic

$$y = ax^2 + bx + c$$

We get, using the linearity of  $\Delta$  and simple algebra,

$$\begin{aligned} \Delta y &= \Delta(ax^2 + bx + c) = a \Delta(x^2) + b \Delta x + c \Delta(1) \\ &= a[(x+h)^2 - x^2] + b[(x+h) - x] + c[1 - 1] \\ &= 2axh + ah^2 + bh \end{aligned}$$

The difference operator applied to a product gives

$$\begin{aligned} \Delta[f(x)g(x)] &= f(x+h)g(x+h) - f(x)g(x) \\ &= f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x) \\ &= f(x+h) \Delta g(x) + g(x) \Delta f(x) \end{aligned}$$

Since  $f(x)$  and  $g(x)$  are interchangeable, this is equivalent to the alternate form

$$\Delta[f(x)g(x)] = f(x) \Delta g(x) + g(x+h) \Delta f(x)$$

Note that the  $x+h$  appears in *only* one argument in either case. Also note how closely the formula resembles that of the calculus

$$\frac{d}{dx}[f(x)g(x)] = f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)]$$



Similarly for the quotient we get

$$\begin{aligned}
 \Delta \left[ \frac{f(x)}{g(x)} \right] &= \frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \\
 &= \frac{f(x+h)g(x) - g(x+h)f(x)}{g(x)g(x+h)} \\
 &= \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - g(x+h)f(x)}{g(x)g(x+h)} \\
 &= \frac{g(x) \Delta f(x) - f(x) \Delta g(x)}{g(x)g(x+h)}
 \end{aligned}$$

which again has one argument at  $x+h$  and closely resembles the corresponding formula from the calculus

$$\frac{d}{dx} \left[ \frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{g^2(x)}$$

Calculus books usually give a long list of formulas for derivatives, both for the direct use and for later inversion to get a table of integrals. Similarly, in the difference calculus we need a list both for present use and for the later inversion into a summation table. We will, however, give only a short list.

$$\begin{aligned}
 \Delta \sin(ax+b) &= 2 \sin \frac{ah}{2} \cos \left[ a \left( x + \frac{h}{2} \right) + b \right] \\
 \Delta \cos(ax+b) &= -2 \sin \frac{ah}{2} \sin \left[ a \left( x + \frac{h}{2} \right) + b \right] \\
 \Delta \tan(ax+b) &= \sin ah \sec(ax+b) \sec[a(x+h)+b] \\
 \Delta \arctan(ax+b) &= \frac{ah}{1 + [a(x+h)+b](ax+b)} \\
 \Delta a^x &= a^x(a^h - 1) \\
 \Delta \ln x &= \ln \left( 1 + \frac{h}{x} \right)
 \end{aligned}$$

In some respects the role of the number  $e$  in the calculus is played by the number 2 in the finite difference calculus. For example, if  $a^h = 2$ , then

$$\Delta a^x = a^x$$

In the common case of unit spacing ( $h = 1$ ) we have  $a = 2$  and

$$\Delta 2^x = 2^x$$

## PROBLEMS

9.2.1 Verify the above list of results.

9.2.2 Show that

$$\Delta\left(\frac{1}{1+x^2}\right) = \frac{-h(2x+h)}{(1+x^2)[1+(x+h)^2]}$$

9.2.3 Show that

$$\Delta \sinh x = \frac{(e^h - 1)(e^x + e^{-(x+h)})}{2}$$

9.2.4 Find

$$\Delta(x \sin x)$$

9.2.5 Find

$$\Delta[\sec^2(ax + b)]$$

9.2.6 Show that

$$\Delta\left(\frac{1}{\sqrt{x}}\right) = \frac{-h}{\sqrt{x}\sqrt{x+h}(\sqrt{x} + \sqrt{x+h})}$$

9.2.7 Show that

$$(a) \quad \Delta(x^4) = \Delta(x^2 \cdot x^2)$$

$$(b) \quad \Delta(x^n) = \Delta(x^{n-1} \cdot x)$$

9.2.8 Show that

$$(a) \quad \Delta \sin(ax + b) = 2 \sin \frac{ah}{2} \sin\left(ax + b + \frac{h}{2} + \frac{\pi}{2}\right)$$

$$(b) \quad \Delta \cos(ax + b) = 2 \sin \frac{ah}{2} \cos\left(ax + b + \frac{h}{2} + \frac{\pi}{2}\right)$$

9.2.9 Find

$$\Delta\left(\frac{ax+b}{cx+d}\right)$$

9.2.10 Find

$$\Delta(2^x \sin x)$$

## 9.3 REPEATED DIFFERENCES

Since  $\Delta f(x)$  is a function of  $x$ , we can apply the  $\Delta$  operator again to obtain the second difference

$$\Delta [\Delta f(x)] \equiv \Delta^2 f(x)$$

This notation corresponds to the notation for the second derivative in the ordinary calculus

$$\frac{d}{dx} \left\{ \frac{d[f(x)]}{dx} \right\} = \frac{d^2}{dx^2} [f(x)]$$

Repeated use of the  $\Delta$  operation gives

$$\Delta[\Delta^{r-1}f(x)] = \Delta^r f(x)$$

In the example of the quadratic (Sec. 9.2)

$$y(x) = ax^2 + bx + c$$

we got

$$\Delta y = 2ahx + ah^2 + bh$$

The second difference is

$$\Delta^2 y = 2ah(x + h) - 2ahx = 2ah^2$$

while the third difference is, clearly,

$$\Delta^3 y = 0$$

It is not an accident that the third difference is zero; rather it is the consequence of the following important theorem.

**The fundamental theorem of the difference calculus** The  $N$ th difference of a polynomial of degree  $N$

$$y(x) = a_N x^N + a_{N-1} x^{N-1} + \cdots + a_0$$

is a constant  $a_N N!/h^N$ , and the  $(N + 1)$ st difference is zero. The proof is very much like that found in the calculus for the corresponding result. We first prove the lemma.

**LEMMA** If  $y(x)$  is a polynomial of degree  $N$ , then  $\Delta y(x)$  is a polynomial of exactly degree  $N - 1$ .

**PROOF OF LEMMA** For the special function  $y(x) = x^N$  we find, using the binomial theorem,

$$\begin{aligned} \Delta y &= (x + h)^N - x^N = \sum_{k=0}^N C(N, k) x^{N-k} h^k - x^N \\ &= Nh x^{N-1} + \frac{N(N-1)}{2} h^2 x^{N-2} + \cdots + h^N \end{aligned}$$

Thus, when the  $\Delta$  operator is applied, the term  $x^N$  becomes a polynomial of exactly degree  $N - 1$  with leading coefficient  $Nh$ . Using the linearity property of the operator decreases each term of a polynomial by one degree, and the term in  $x^{N-1}$  cannot cancel out. Thus the lemma is proved. ////

**PROOF** We simply use the lemma  $N$  times and find that only the term from  $x^N$  remains and that it has the coefficient  $N!/h^N$ . The next difference makes this constant cancel, and the theorem is proved. ////

This theorem is basic to much of the classical parts of numerical analysis; in one way or another an appeal is regularly made to it.

### PROBLEMS

9.3.1 Prove the theorem by mathematical induction.

9.3.2 Using  $h = 1$ , compute the second and fourth differences of  $y = x^4 - 4x^3 + 6x^2 - 4x + 1$ . *Ans.*  $\Delta^2 y = 12(x-1)^2$ ,  $\Delta^4 y = 24$

9.3.3 Using  $h = 1$ , find all the differences of  $x(x-1)(x-2)(x-3)$ .

9.3.4 Find

(a)  $\Delta^k \left( \frac{1}{x} \right)$

(b)  $\Delta^k (\ln x)$

9.3.5 Find

$$\Delta^k [\sin(a + bx)]$$

$$\Delta^k [\cos(a + bx)]$$

*Hint:* Prob. 9.2.8

9.3.6 Find  $\Delta^k(a^x)$ .

### 9.4 THE DIFFERENCE TABLE

When using higher differences, it is useful to imagine that the numbers are arranged in the form of a difference table (Table 9.4.1), although the differences are probably not stored in a machine in this way.

Table 9.4.1 DIFFERENCE TABLE

| $x$ | $y(x)$ | $\Delta y(x)$ | $\Delta^2 y(x)$ | $\Delta^3 y(x)$ |
|-----|--------|---------------|-----------------|-----------------|
| 0   | $y(0)$ |               |                 |                 |
| 1   | $y(1)$ | $\Delta y(0)$ |                 |                 |
| 2   | $y(2)$ | $\Delta y(1)$ | $\Delta^2 y(0)$ |                 |
| 3   | $y(3)$ | $\Delta y(2)$ | $\Delta^2 y(1)$ | $\Delta^3 y(0)$ |
| 4   | $y(4)$ | $\Delta y(3)$ | $\Delta^2 y(2)$ | $\Delta^3 y(1)$ |
| 5   | $y(5)$ | $\Delta y(4)$ | $\Delta^2 y(3)$ | $\Delta^3 y(2)$ |
| .   | .      |               |                 |                 |
| .   | .      |               |                 |                 |
| .   | .      |               |                 |                 |

As an example of a difference table consider the values of the sine-integral function as shown in Table 9.4.2 (we have used the common convention and have written the differences as if the decimal point were at the right-hand end of the numbers). Higher differences are usually carried until the numbers either tend to oscillate or are mainly zero.

It is easy to check a difference table for accuracy (when computed by hand). If we sum any column of differences and add the sum to the top number in the preceding column, we must get the bottom number in the preceding column. For example, in Table 9.4.2 the sum of the  $\Delta^3$  is  $-26$ . This added to the  $-2$  of  $\Delta^2(0)$  gives  $-2 - 26 = -28$ , the bottom value of the  $\Delta^2$  column, namely  $\Delta^2(0.8)$ .

Most tables are given in fixed-point notation, at least locally, though over various ranges differing numbers of figures may be printed. This means that it is often possible to reconstruct the table from the top diagonal line of differences plus the last column of differences. In a sense the differences contain the same information that the function does. The only exception to this is when the function, or some column of differences, changes sign and the difference (which is now a sum of two numbers) in the next column requires a shift and possible loss of information due to the carry beyond the extreme left.

Table 9.4.2 DIFFERENCE TABLE OF

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt$$

| $x$ | $\text{Si}(x)$ | $\Delta$ | $\Delta^2$ | $\Delta^3$ | $\Delta^4$ |
|-----|----------------|----------|------------|------------|------------|
| 0.0 | 0.0000         |          |            |            |            |
|     |                | 999      |            |            |            |
| 0.1 | 0.0999         |          | -2         |            |            |
|     |                | 997      |            | -6         |            |
| 0.2 | 0.1996         |          | -8         |            | 5          |
|     |                | 989      |            | -1         |            |
| 0.3 | 0.2985         |          | -9         |            | -4         |
|     |                | 980      |            | -5         |            |
| 0.4 | 0.3965         |          | -14        |            | 3          |
|     |                | 966      |            | -2         |            |
| 0.5 | 0.4931         |          | -16        |            | -1         |
|     |                | 950      |            | -3         |            |
| 0.6 | 0.5881         |          | -19        |            | 0          |
|     |                | 931      |            | -3         |            |
| 0.7 | 0.6812         |          | -22        |            | 0          |
|     |                | 909      |            | -3         |            |
| 0.8 | 0.7721         |          | -25        |            | 0          |
|     |                | 884      |            | -3         |            |
| 0.9 | 0.8605         |          | -28        |            |            |
|     |                | 856      |            |            |            |
| 1.0 | 0.9461         |          |            |            |            |

Although we have used the tables as if they started at zero, it is frequently useful to imagine that the table extends indefinitely in both directions and we have only a sample of the entire function. At other times it is useful to shift the origin to the place in which we are currently interested. Thus we will be quite vague about the actual origin of a table and occasionally shift without mentioning it specifically.

### PROBLEMS

- 9.4.1 Make a difference table for the sine function at  $10^\circ$  spacing, using a 5-place table ( $0 \leq x \leq 90^\circ$ ). Check your arithmetic by the above mentioned method.
- 9.4.2 Make a difference table for  $y = x^4$ .
- 9.4.3 Describe a difference table for the function  $y = a^x$ .
- 9.4.4 Describe a difference table for the function  $y = \sin(\pi x/3)$ .

## 9.5 TABULATING A POLYNOMIAL AT A REGULAR SPACING

As observed in Sec. 3.10, we frequently wish to evaluate a function at a sequence of equally spaced points. This is especially true in evaluating polynomials. For example, it is customary in processing data to first remove a polynomial trend. This means that the polynomial must be evaluated at every point so that it can be subtracted from the data.

The difference table gives a powerful way of constructing the left-hand column from the top line of differences plus one column of high-order differences. For a polynomial we know from the fundamental theorem that the  $N$ th differences are a constant, and so the top line of differences alone will enable us to reconstruct the whole table. The process is as follows and is based on the obvious relation: From

$$\Delta f(k) = f(k+1) - f(k)$$

we get

$$f(k+1) = f(k) + \Delta f(k)$$

Consider the specific example of

$$\begin{array}{ll} y(x) = 3x^2 - 6x + 9 & y(0) = 9 \\ \Delta y(x) = & 6x - 3 \quad \Delta y(0) = -3 \\ \Delta^2 y(x) = 6 & \Delta^2 y(0) = 6 \end{array}$$

Table 9.5 shows the method of construction from the differences.

Even when a function is not exactly a polynomial, this still provides a powerful method of constructing many values, before making a small adjustment due to the fact that the differences are not exactly a constant.

However, roundoff is a problem, as always. In the case of a polynomial if the calculation of the differences is exact, then the rest of the arithmetic can usually be kept under reasonable control. But if there is a small roundoff error in the differences, this error will be greatly magnified in the function. For example, consider the table with all zeros to be true, but suppose that the  $k$ th differences are in error by a small fixed amount  $\varepsilon$  at each location. What does the function become?

The preceding column becomes  $n\varepsilon$ , the next  $[n(n-1)/2]\varepsilon, \dots$ , etc. Roughly the function looks like

$$\frac{n^k}{k!} \varepsilon$$

which in time becomes very large. Thus it is customary, when using this method of computing a function, to periodically correct the accumulated errors.

Table 9.5  $y(x) = 3x^2 - 6x + 9$

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ |
|-----|-----|------------|--------------|
| 0   | 9   |            |              |
| 1   | 6   | -3         |              |
| 2   | 9   | 3          | 6            |
| 3   | 18  | 9          | 6            |
| 4   | 33  | 15         | 6            |
| 5   | 54  | 21         | 6            |
| 6   | 81  | 27         | 6            |
| 7   | 114 | 33         | 6            |
| 8   | 153 | 39         | 6            |
| 9   | 198 | 45         | 6            |
| 10  | 249 | 51         |              |

Check:  $y(10) = 300 - 60 + 9 = 249$

## 9.6 THE FACTORIAL NOTATION

In the calculus  $x^n$  plays an important role. For example, in the Taylor series

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots = \sum_{k=0}^{\infty} a_k x^k$$

an arbitrary function is expanded in powers of  $x$ . The main reason that  $x^n$  plays this leading role is that

$$\frac{d}{dx} x^n = nx^{n-1} \quad n \geq 1$$

Thus the coefficients of the Taylor series are determined by differentiating and evaluating the derivatives at the origin

$$\begin{aligned} f(0) &= a_0 \\ f'(x) &= a_1 + 2a_2 x + 3a_3 x^2 + \cdots \\ f'(0) &= a_1 \\ f''(x) &= 2a_2 + 6a_3 x + \cdots \\ f''(0) &= 2a_2 \end{aligned}$$

and in general

$$f^{(k)}(0) = k!a_k$$

When differences are used instead of derivatives, the functions corresponding to  $x^n$  are

$$g_n(x) \equiv x(x-1) \cdots (x-n+1)$$

where there are  $n$  factors.

We have

$$\begin{aligned} \Delta g_n(x) &= (x+1)x(x-1) \cdots (x-n+2) - x(x-1) \cdots (x-n+1) \\ &= [(x+1) - (x-n+1)]x(x-1) \cdots (x-n+2) \\ &= nx(x-1) \cdots (x-n+2) \end{aligned}$$

For convenience we will write

$$g_n(x) = x^{(n)} \equiv x(x-1)(x-2) \cdots (x-n+1) \quad n = 1, 2, \dots$$

and we have just shown that

$$\Delta x^{(n)} = nx^{(n-1)}$$

We need to extend the range of definition to all integers (much as were the powers of  $x$  in algebra). As a basis we use the identity

$$x^{(n)} = x^{(m)}(x-m)^{(n-m)} \quad n > m \geq 1$$



If we formally set  $m = 0$ , we get

$$x^{(n)} = x^{(0)} x^{(n)}$$

which suggests that we define

$$x^{(0)} \equiv 1$$

much as we made  $0! \equiv 1$ .

For negative exponents we use the same identity and set  $n = 0$

$$1 = x^{(0)} = x^{(m)} (x - m)^{(-m)}$$

or

$$(x - m)^{(-m)} = \frac{1}{x^{(m)}}$$

Set  $x - m = y$  and we have the more convenient form

$$y^{(-m)} = \frac{1}{(y + m)^{(m)}} = \frac{1}{(y + m)(y + m - 1) \cdots (y + 1)}$$

Note that

$$x^{(m)} x^{(-m)} \neq x^{(0)} \quad m \neq 0$$

Corresponding to the Taylor-series expansion in powers of  $x$ , we have a formal expansion in factorials

$$f(x) = b_0 + b_1 x^{(1)} + b_2 x^{(2)} + \cdots = \sum_{k=0}^{\infty} b_k x^{(k)}$$

The coefficients are determined in exactly the same way, this time by differencing and evaluating at zero

$$\begin{aligned} f(0) &= b_0 \\ \Delta f(0) &= b_1 \\ \Delta^2 f(0) &= 2b_2 \\ &\dots\dots\dots \\ \Delta^k f(0) &= k! b_k \end{aligned}$$

Hence we have the *Newton expansion*

$$f(x) = \sum_{k=0}^{\infty} \frac{\Delta^k f(0)}{k!} x^{(k)} = \sum \Delta^k f(0) C(x, k)$$

where  $C(x, k)$  is the usual binomial coefficient.

The conversion of a polynomial expressed in powers of  $x$  to a sum of factorials is straightforward and is based on the same idea used to convert: a

number of a given base (Sec. 2.7); a polynomial to a sum of powers of  $x - a$  (Sec. 3.7); or a polynomial to powers of a quadratic (Sec. 6.7). We divide the polynomial by  $x$ , the quotient by  $x - 1$ , that quotient by  $x - 2$ , etc. The successive remainders are the coefficients of the factorial representation. For example, consider the polynomial

$$P_4(x) = 2x^4 - 3x^3 + x^2 - 1$$

|   |   |    |    |   |    |    |
|---|---|----|----|---|----|----|
| 0 | 2 | -3 | 1  | 0 | -1 |    |
|   |   | 0  | 0  | 0 | 0  |    |
| 1 | 2 | -3 | 1  | 0 | -1 | -1 |
|   |   | 2  | -1 | 0 |    |    |
| 2 | 2 | -1 | 0  | 0 |    | 0  |
|   |   | 4  | 6  |   |    |    |
| 3 | 2 | 3  | 6  |   |    | 6  |
|   |   | 6  |    |   |    |    |
|   | 2 | 9  |    |   |    |    |

then

$$P_4(x) = 2x^{(4)} + 9x^{(3)} + 6x^{(2)} + 0x^{(1)} - 1$$

The first factor by zero need not be done, of course, but was included for logical presentation.

### PROBLEMS

9.6.1 Using Newton's formula, find a polynomial which takes on the following values of  $P(n)$ .

|        |    |    |    |    |    |    |
|--------|----|----|----|----|----|----|
| $n$    | 0  | 1  | 2  | 3  | 4  | 5  |
| $P(n)$ | 41 | 43 | 47 | 53 | 61 | 71 |

*Ans.*  $P(n) = n^2 + n + 41$  [for  $n < 41$  note that  $P(n)$  = prime number]

9.6.2 Show that

$$1^{(-m)} = \frac{1}{(m+1)!}$$

9.6.3 Show that

$$n^{(n)} = n!$$

9.6.4 Compute  $x^{(m)}x^{(-m)}$  where  $x = m$ .

*Ans.*  $\frac{(m!)^2}{(2m)!}$

9.6.5 Simplify  $1/x^{(-m)}$ .

9.6.6 Describe the difference table of  $x^{(k)}$ .

**\*9.7 STIRLING NUMBERS OF THE FIRST KIND**

Although we have already shown how to go from a polynomial in powers of  $x$  to its factorial representation, the central role of powers of  $x$  in the continuous case and of factorials in the discrete case requires that we look more closely at the topic.

To express  $x^{(n)}$  in powers of  $x$  we write

$$x^{(n)} = \sum_{k=0}^n s(n, k)x^k \quad (9.7.1)$$

and set out to find the numbers  $s(n, k)$ . These numbers are called *Stirling<sup>1</sup> numbers of the first kind*. For  $n = 1$  we have

$$x^{(1)} = x = s(1, 0) + s(1, 1)x$$

whence

$$s(1, 0) = 0 \quad s(1, 1) = 1$$

For  $n = 2$  we have

$$x(x - 1) = x^2 - x = s(2, 0) + s(2, 1)x + s(2, 2)x^2$$

whence

$$s(2, 0) = 0 \quad s(2, 1) = -1 \quad s(2, 2) = 1$$

It is easy to see that for all  $n \geq 1$ ,  $s(n, 0) = 0$ , but  $s(0, 0) = 1$ .

Rather than continuing one power at a time we write

$$x^{(n+1)} = (x - n)x^{(n)}$$

and use Eq. (9.7.1) on both sides to get

$$\begin{aligned} \sum_{k=1}^{n+1} s(n+1, k)x^k &\equiv (x - n) \sum_{k=1}^n s(n, k)x^k \\ &\equiv \sum_{k=1}^n s(n, k)x^{k+1} - n \sum_{k=1}^n s(n, k)x^k \\ &\equiv \sum_{k=1}^n [s(n, k-1) - ns(n, k)]x^k + s(n, n)x^{n+1} \end{aligned}$$

Table 9.7

| $n \backslash k$ | 1  | 2   | 3  | 4   | 5 |
|------------------|----|-----|----|-----|---|
| 1                | 1  |     |    |     |   |
| 2                | -1 | 1   |    |     |   |
| 3                | 2  | -3  | 1  |     |   |
| 4                | -6 | 11  | -6 | 1   |   |
| 5                | 24 | -50 | 35 | -10 | 1 |

<sup>1</sup> James Stirling (1692–1770).

Equating the highest power of  $x$ , we get

$$s(n+1, n+1) = s(n, n) = \cdots = s(1, 1) = 1 = s(0, 0)$$

For  $x^k$  we get

$$s(n+1, k) = s(n, k-1) - ns(n, k) \quad k = 1, 2, \dots, n$$

There is no simple formula for the Stirling numbers of the first kind. Table 9.7 gives a few of the numbers.

### PROBLEMS

9.7.1 Extend Table 9.7 one line to  $n = 6$ .

9.7.2 Using Table 9.7, show that  $x^{(3)} + 2x^{(2)} + x^{(1)} - 1 = (x^2 + 1)(x - 1)$ .

9.7.3 Using Table 9.7, write  $x^{(5)} + 10x^{(4)}$  as a polynomial.

### \*9.8 STIRLING NUMBERS OF THE SECOND KIND

The Stirling numbers of the second kind express  $x^n$  in terms of factorials:

$$x^n = \sum_{k=0}^n S(n, k)x^{(k)}$$

As in Sec. 9.7, we first compute a few cases and then find the general relation. For  $n = 1$

$$x = S(1, 0) + S(1, 1)x$$

so that

$$S(1, 0) = 0 \quad S(1, 1) = 1$$

For  $n = 2$

$$x^2 = S(2, 0) + S(2, 1)x + S(2, 2)x(x-1)$$

so that

$$S(2, 0) = 0 \quad S(2, 1) = 1 \quad S(2, 2) = 1$$

Table 9.8

| $n \backslash k$ | 1 | 2  | 3  | 4  | 5 |
|------------------|---|----|----|----|---|
| 1                | 1 |    |    |    |   |
| 2                | 1 | 1  |    |    |   |
| 3                | 1 | 3  | 1  |    |   |
| 4                | 1 | 7  | 6  | 1  |   |
| 5                | 1 | 15 | 25 | 10 | 1 |

Again it is clear that

$$S(n, 0) = 0 \quad n > 0$$

and

$$S(0, 0) = 1$$

The recurrence relation follows from

$$\begin{aligned} x^{n+1} &= x \cdot x^n \\ \sum_{k=0}^{n+1} S(n+1, k)x^{(k)} &= x \sum_{k=0}^n S(n, k)x^{(k)} = \sum_{k=0}^n S(n, k)(x-k)x^{(k)} + \sum_{k=0}^n kS(n, k)x^{(k)} \\ &= \sum_{k=0}^n S(n, k)x^{(k+1)} + \sum_{k=0}^n kS(n, k)x^{(k)} \\ &= \sum_{k=0}^n [S(n, k-1) + kS(n, k)]x^{(k)} + S(n, n)x^{(n+1)} \end{aligned}$$

Equate like factorials

$$S(n+1, k) = S(n, k-1) + kS(n, k) \quad k = 1, 2, \dots, n$$

$$S(n+1, n+1) = S(n, n) = \dots = S(1, 1) = 1 = S(0, 0)$$

Some of the numbers are given in Table 9.8.

## PROBLEMS

9.8.1 Extend Table 9.8 to  $n = 6$ .

9.8.2 Write  $x^3$  in factorial form using Table 9.8. Check by synthetic division.

9.8.3 Write  $x^3 + 7x^2 - 9x + 3$  in factorial form.

## 9.9 ALTERNATE NOTATIONS

The choice of the notation for the first difference

$$\Delta f(x) = f(x+h) - f(x)$$

was arbitrary and unfortunately not symmetric. We might just as well have chosen the *backward* differences

$$\nabla f(x) = f(x) - f(x-h)$$

What would have happened if we had? We would have found that the ascending factorials

$${}^{(n)}x = x(x+h)(x+2h) \cdots [x+(n-1)h]$$

would give the relation we need for handling things, and we would have found a *backward* Newton formula. Similarly we would have found different Stirling numbers  $\sum(n, k)$  and  $\sigma(n, k)$ . These are related to the old ones by

$$\begin{aligned}\sum(n, k) &= (-1)^{n+k} s(n, k) = |s(n, k)| \\ \sigma(n, k) &= (-1)^{n+k} S(n, k)\end{aligned}$$

Thus, there is a completely analogous theory for the backward differences. The resulting formulas are occasionally useful, but in a first course they are a luxury.

Similarly *central differences* and *mean differences* are sometimes used. Again they are merely a change in notation and a convenience in some situations but not worth the trouble in an introductory course.

Many books make use of elaborate symbolic methods, for which the alternate notations are quite useful. We have used only the operator  $\Delta$  in a few simple, finite cases and have so far avoided the more dubious symbolic operators. For example, the Taylor series can be written

$$\begin{aligned}f(a+h) &= f(a) + hf'(a) + h^2 f''(a) + \cdots \\ &= \sum_{k=0}^{\infty} \frac{h^k D^k}{k!} [f(a)] \\ &= e^{hD} f(a)\end{aligned}$$

Therefore

$$\Delta f(a) = (e^{hD} - 1)f(a)$$

or symbolically

$$\begin{aligned}\Delta &= e^{hD} - 1 \\ e^{hD} &= 1 + \Delta \\ hD &= \ln(1 + \Delta) = \Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \cdots \\ D &= \frac{1}{h} \left( \Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \cdots \right)\end{aligned}$$

We will, however, occasionally do some formal manipulation with what amounts to symbolic methods, because the methods lead easily and rapidly to useful results. The history of mathematics is full of processes that were once condemned and are now considered legitimate—but also of ones that lead to false results. It is an art to be safe but not rigid.

### 9.10 AN EXAMPLE OF AN INTEGRAL EQUATION

With even this little bit of theory it is time to give an example of how it can be used to solve useful problems. The illustration is one that occurred in solid-state physics. The problem is to calculate

$$g(y) = \frac{d}{dy} \int_0^y \frac{f(x)}{\sqrt{y-x}} dx \quad 0 \leq y \leq 1$$

when the data for  $f(x)$  is given at  $x = 0, 0.1, 0.2, \dots, 1.0$ .

Examining the problem, we immediately see that at the upper limit, when  $x = y$ , the integrand becomes infinite. Furthermore, the result of the integration is going to be differentiated with respect to  $y$ , and when we differentiate under the integral sign with respect to  $y$ , we will get an integral that diverges. Thus the problem requires some care.

To start our thinking, for what functions  $f(x)$  could we solve the problem? It is natural to try  $x^n$

$$g(y) = \frac{d}{dy} \int_0^y \frac{x^n}{\sqrt{y-x}} dx$$

This suggests the usual substitution

$$x = y \sin^2 \theta$$

$$dx = 2y \sin \theta \cos \theta d\theta$$

and

$$g(y) = \frac{d}{dy} \left( \frac{2y^{n+1}}{\sqrt{y}} \right) \int_0^{\pi/2} \sin^{2n+1} \theta d\theta = \frac{(2n+1)y^n}{\sqrt{y}} W_{2n+1}$$

where  $W_{2n+1}$  is the Wallis<sup>1</sup> integral

$$\begin{aligned} W_{2n+1} &= \int_0^{\pi/2} \sin^{2n+1} \theta d\theta = \frac{2 \cdot 4 \cdots (2n)}{1 \cdot 3 \cdot 5 \cdots (2n+1)} \\ &= \frac{2n}{2n+1} W_{2n-1} \quad W_1 = 1 \end{aligned}$$

Thus we can handle all the cases of  $x^n$ . Since the operator

$$\frac{d}{dy} \int_0^y \frac{(\cdot)}{\sqrt{y-x}} dx$$

is linear, if we can write

$$f(x) = a_0 + a_1 x + \cdots + a_{10} x^{10} = \sum_{k=0}^{10} a_k x^k$$

<sup>1</sup> John Wallis (1616–1703).

we will have as the solution

$$g(y) = \frac{1}{\sqrt{y}} \sum_{n=0}^{10} (2n+1) W_{2n+1} a_n y^n$$

To find the representation of  $f(x)$  we use Newton's formula (with a spacing of  $\frac{1}{10}$ )

$$\begin{aligned} f(x) = f(0) + 10x \Delta f(0) + \frac{10x(10x-1)}{2!} \Delta^2 f(0) \\ + \cdots + (10x)^{(10)} \Delta^{10} f(0) \end{aligned}$$

The Stirling numbers of the first kind will get us from the factorial expansion to the polynomial we want, and we are done.

As we will later see, it is often dangerous to use a polynomial of degree as high as 10, but in the case that inspired the example, where the data were crude and the answer was not expected to be much better, the results were satisfactory as judged by these facts:

- 1 The graph of the approximating polynomial we found from Newton's formula looked reasonable.
- 2 The results were compatible with other parts of the physical theory.
- 3 The results stimulated further laboratory measurements of the same effect.

As a final step in the problem, why did we succeed? What other kinds of problems can we handle the same way? Evidently if we can analytically integrate for some family of functions and can represent our function as a linear combination of the family, then we have a solution.



# 10

## ROUND OFF ESTIMATION

### 10.1 WHY ROUND OFF AGAIN?

By now the reader has begun to suspect that numerical methods are much too concerned with roundoff. After all it is a small effect and should not bother the practical man who needs three or four figures at most. If single precision is not enough, then double precision surely will do the trick, and modern machines often have double precision built in so that it is hardly slower than single precision.

But consider the following situation. You are using the zeros of a polynomial, and the output from the library routine (unlike the one proposed in Chap. 6) produces, due to roundoff, a pair of close zeros instead of a double zero. As we discussed in Sec. 6.1, this will probably lead to severe cancellation in the subsequent computing. When the problem is repeated with double precision, the zeros reported by the polynomial routine are, of course, just that much closer, and the subsequent computation again loses almost all the accuracy there is. Double precision did not get you out of the trouble as it should have; it merely moved it around a bit and made it look a bit different.

It is often true, as many people say, that double precision is the answer to roundoff troubles. At least most of the time this is true *provided* you know that

roundoff has caused the trouble. But among other questions that naturally arise is, how are you to know *after* a computation is done that roundoff has caused inaccuracies? By looking at the printed answers? Just how are you going to do this and make any reasonable estimate of the roundoff present? One of the goals of this chapter is to answer this question when the results are in the form of a table of numbers at equal spacing of some variable or parameter.

It is also reasonable to ask for an estimate of the roundoff *before* a computation is begun. Of course this estimate will be less accurate than one made after the computation is done, but it also may be more valuable in saving machine and human time. We have maintained that it is better to avoid serious roundoff by using forethought than to bound it after it happens. It is further reasonable to ask how the machine can help us with the problem of roundoff *during* the computation.

Looking at the problem another way, there are (1) estimates of bounds, (2) guaranteed bounds, (3) statistical estimates of roundoff and its variance, and (4) even backward analysis (mentioned in Secs. 3.8 and 7.10), which gives answers to roundoff in the form "The answer you obtained is the exact answer to a problem that is within so much of the original problem."

This whole chapter is devoted to a few aspects of roundoff. We will treat roundoff again from an entirely different point of view in Part III—a viewpoint which in many respects is more fundamental than this first careful look at a difficult, important topic. Unfortunately for the user, all too little is known about roundoff. When he faces his first large decision based on extensive numerical computation, the user will begin to see why he wishes he knew more about it.

## 10.2 RANGE ARITHMETIC (INTERVAL ARITHMETIC<sup>1</sup>)

The basic idea of range, or interval, arithmetic is very simple. In place of each number we carry two numbers, the largest (maximum) and the least (minimum) that it could be. At the start of a computation these ranges are assigned, perhaps of zero length. Each arithmetic operation is replaced by a pair of operations, one that finds the largest and rounds this up and the other that finds the smallest and rounds this down. Thus, the result of each arithmetic operation is a pair of numbers marking the range in which it is *certain* that the result lies.

Range arithmetic is usually simulated by an interpretive system on a machine, though in principle it could be wired into a machine. The details of the simulation are messy but not fundamentally difficult.

<sup>1</sup> See Moore [43].

In a short computation the resulting range is usually reasonable in size, but in a long computation the range may be so great that it is useless. It may be thought that the true answer lies near the middle of the range, but this is clearly false since if the range of  $x$  were in the interval  $0 \leq x \leq 1$  and the same were true for  $y$ , then the assumption that both lie near the middle leads to the product lying near the  $1/4$  mark in the range, which is not the middle of its range.

Range arithmetic has another fault. It is the essence of good, iterative computation that the iterations improve the result, but that the improvement takes place at no one step in the loop. For example, in the trivial case of applying Newton's method for finding the square root of a number  $N$ , we are led to

$$\begin{aligned} y &= x^2 - N \\ y' &= 2x \\ x_{n+1} &= x_n - \frac{x_n^2 - N}{2x_n} \\ &= \frac{1}{2} \left( x_n + \frac{N}{x_n} \right) \end{aligned}$$

In words, the new guess is the average of the current estimate  $x_n$  and the quotient  $N/x_n$ . If one number is too large, then the other is automatically too small, and the average is a very good estimate. How is range arithmetic to sense that this simple loop has in it this internal correlation between the two numbers? Yet this is the essence of good computing.

All that range arithmetic can do is provide some bounds, probably very much too wide, within which the result must lie. Sometimes this is good enough and the above objections are irrelevant, but sometimes they are very appropriate. Thus range arithmetic is a valuable (at times expensive) tool, but it is not the complete answer to roundoff.

## PROBLEMS

- 10.2.1 Draw a flow diagram for the operation of multiplication in simulating range arithmetic. (Note a range may cover 0.)
- 10.2.2 Do the same as in Prob. 10.2.1 but for division.
- 10.2.3 Discuss the characteristics of a loop in which the range will correctly contract as the iterations progress.

### 10.3 ERROR PROPAGATION IN A DIFFERENCE TABLE

The difference table, Table 9.4.1, is one of the basic tools of roundoff estimation. Suppose we have a table of equally spaced values of a function and there is one error and this is at the  $i$ th entry,  $y_i(1 + \varepsilon_i)$ . What happens in the difference table? The difference operator is linear, and so the resulting difference table can be regarded as the sum of the correct values plus the difference table of the single error  $y_i \varepsilon_i$ . This latter table is of the form shown in Table 10.3 (where we have suppressed the multiplicative factor  $y_i \varepsilon_i$  from all the numbers).

The numbers in the table are the binomial coefficients. To prove this we merely write the operator equation, where  $E$  is the displacement operator,  $Ef(x) = f(x + h)$ . Since

$$\Delta[f(x)] = f(x + h) - f(x)$$

we have

$$\Delta = E - 1$$

and

$$\begin{aligned}\Delta^k &= (E - 1)^k \\ &= \sum (-1)^m C(k, m) E^{k-m}\end{aligned}$$

Table 10.3

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ | $\Delta^5 y$ | $\Delta^6 y$ |
|-----|-----|------------|--------------|--------------|--------------|--------------|--------------|
| .   | .   | .          | .            | .            | .            | .            | .            |
| .   | .   | .          | .            | .            | .            | .            | .            |
| .   | .   | .          | .            | .            | .            | .            | .            |
| -3  | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| -2  | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| -1  | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 0   | 1   | 1          | -2           | 3            | -4           | 10           | -20          |
| 1   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 2   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 3   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 4   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 5   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |
| 6   | 0   | 0          | 0            | 0            | 0            | 0            | 0            |